

Trauma, Mobile, and Surgical Radiography

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CONTENTS

Trauma and Mobile Imaging

Introduction, 568

Skeletal Trauma and Fracture Terminology, 568

Mobile X-Ray Equipment, 572

Digital Imaging Considerations, 573

Positioning Principles for Trauma and Mobile Radiography, 574

Alternative Modalities, 576

Routine and Special Projections

Trauma and Mobile Positioning, 577

AP Supine and Decubitus: Abdomen, 579

Upper Limb, 581

Shoulder, 584

Lower Limb, 586

Lower Limb Positioning Examples, 587

Pelvis, 589

HIP, 590

Cervical Spine, 591

Thoracic and Lumbar Spine, 592

Cranium—Lateral, Horizontal Beam, 593

Cranium—AP, AP Axial 15° (Reverse Caldwell Method), 594

Cranium—AP 30° Axial (Towne Method), 595

Facial Bones—Lateral, Horizontal Beam, 596

Facial Bones—Acanthioparietal (Reverse Waters Method) and Modified Acanthioparietal (Modified Reverse Waters), 597

Surgical Radiography

Essential Attributes of the Radiologic Technologist in Surgical Radiography, 598

Surgical Team, 599

Surgical Radiography Imaging Equipment, 600

Radiation Protection in Surgical Radiography, 602

Patient Dose, 603

Summary of Radiation Protection in Surgical Radiography, 603

Surgical Asepsis Principles, 605

Surgical Suite Environment, 605

Surgical Attire, 607

Surgical Radiographic Procedures

Biliary Tract Procedures, 608

Urinary Tract Procedures, 610

Orthopedic Procedures, 611

Spinal Procedures, 614

Thoracic Procedures, 616

Glossary of Surgical Abbreviations, Terminology, and Procedures, 617

TRAUMA AND MOBILE IMAGING

Introduction

This chapter is divided into two primary sections; the first section involves **trauma** and **mobile imaging**, and the second section discusses **surgical radiography**. Student radiographers may feel anxious and intimidated by these types of procedures because of the nature of the patient's condition, different equipment, unfamiliar surroundings, and working with personnel outside the radiology department. All of these factors can create a heightened perception of stress. To build confidence, reduce perceived stress, and perform these advanced exams with greater accuracy, it is important to view these experiences as an opportunity to build on knowledge previously acquired. There are multiple facets to these areas that could be discussed in great detail; however, the purpose of this chapter is to provide a basic foundation. Experience is truly the greatest resource in becoming prepared to face the challenges presented by these procedures. Maintaining an open mind, thinking critically, and accepting that learning is a product of effort and experience will allow for the development of the skills necessary to be proficient in trauma, mobile, and surgical radiography.

Skeletal Trauma and Fracture Terminology

The American Registry of Radiologic Technologists (ARRT) defines trauma as a serious injury or shock to the body, often requiring modifications that may include variations in positioning, minimal movement of the body part, and so on¹ (Fig. 15.1). This may mean that patients cannot be brought to the radiology department for routine radiographic procedures as described in other sections of this text. Instead, a mobile (portable) x-ray unit must be taken to the emergency department (ED) or to the patient's bedside (Fig. 15.2). Even if patients are brought to the radiology department, they may be immobilized in a number of ways. Some may present with one or more splints, indicating possible limb fractures or dislocations. Others may be strapped to a backboard with a cervical collar in place. In these cases, **a major adaptation of CR angles and image receptor placement is required**. Radiographers must use their knowledge of anatomy, technical factors, and positioning to acquire diagnostic images in difficult circumstances.

Skeletal trauma and surgical radiography require an understanding of terms that are unique to these situations, such as fracture-dislocation terminology. Knowing the terms used in patient histories or on examination requisitions allows the technologist to understand which type of injury or fracture is suspected and which projections are most important. It also is useful for knowing how to avoid positioning techniques or body positions that may result in additional pain or injury.

DISLOCATION

Dislocation refers to the displacement of a bone that is no longer in contact with its normal articulation.² Dislocations can frequently be clinically identified by the abnormal shape or alignment of the body parts. Any movement of these parts can be painful and must be avoided. As with fractures, dislocations should be imaged in two planes, 90° to each other, to demonstrate the degree of displacement. The most common dislocations encountered in trauma involve the shoulder (Fig. 15.3), fingers or thumb, patella, and hip.

If a bone has relocated itself following the injury, damage may still have occurred, and a minimum of two projections of the affected joint is required to assess for damage and/or possible avulsion fractures.



Fig. 15.1 Trauma radiography.

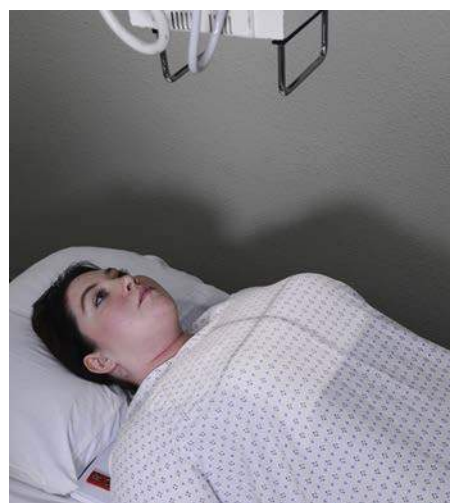


Fig. 15.2 Bedside mobile radiography.

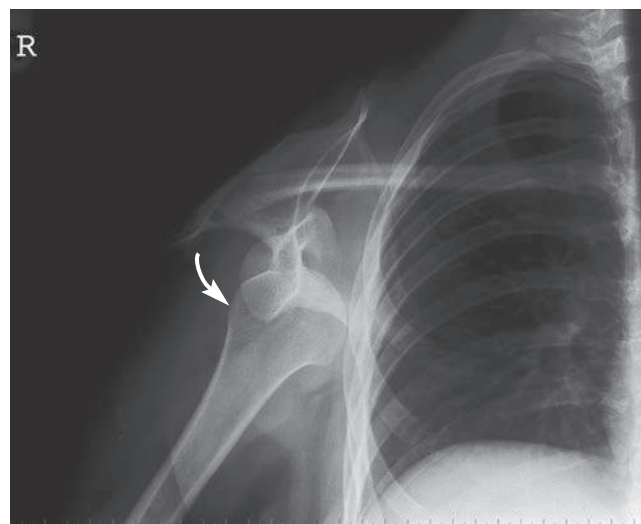


Fig. 15.3 Right shoulder dislocation (AP projection).

SUBLUXATION

A partial dislocation is illustrated in Fig. 15.4, in which a vertebra is displaced posteriorly. Another example is nursemaid elbow (so-called jerked elbow), which is a traumatic partial dislocation of the radial head of a child. This is caused by a hard pull on the hand and wrist of a child by an adult. It is frequently reduced when the forearm is supinated for an AP elbow projection.

SPRAIN

A sprain is a forced wrenching or twisting of a joint resulting in a partial rupture or tearing of supporting ligaments, without dislocation. A sprain may result in severe damage to associated blood vessels, tendons, ligaments, and/or nerves. Severe swelling and discoloration caused by hemorrhage of ruptured blood vessels frequently accompany a severe sprain. A severe sprain can be painful and must be handled with great care during the radiographic examination. Symptoms are similar to those of fractures; radiographs aid in differentiating a sprain from a fracture.

CONTUSION

This is a bruise type of injury with a possible avulsion fracture. An example is a hip pointer, a football injury involving contusion of bone at the iliac crest of the pelvis.

FRACTURE

A fracture (fx) is defined as a disruption of bone caused by mechanical forces applied either directly to the bone or transmitted along the shaft of the bone.² With any possible fracture, the technologist must use extreme caution in moving and positioning the patient so as not to cause further injury or displacement of fracture fragments. The technologist should **never force a limb or body part into position**. If the fracture is obvious, or if severe pain accompanies any movement, positioning should be adapted as needed.

FRACTURE ALIGNMENT TERMINOLOGY

Alignment refers to the associative relationship between long axes of the fracture fragments. A fracture is aligned if the long axes of the bone remain parallel to each other.

Apposition

Apposition describes how the fragmented ends of the bone make contact with each other. Three types of apposition are known:

1. **Anatomic apposition:** Anatomic alignment of ends of fractured bone fragments, wherein the ends of the fragments make end-to-end contact.
2. **Lack of apposition (distraction):** The ends of fragments are aligned but pulled apart and are not making contact with each other (e.g., as from excessive traction; Fig. 15.5).
3. **Bayonet apposition:** The fracture fragments overlap and the shafts make contact, but not at the fracture ends (Fig. 15.6).

Angulation

Angulation describes loss of alignment of the fracture; apex is the direction of the angulation and is opposite in relation to the distal part of the fracture fragments (Fig. 15.7). The following three terms can be used to describe the type or direction of angulation, which uses the apex or distal fragments as its reference point:

1. **Apex angulation:** Describes the direction or angle of the apex of the fracture, such as a medial or lateral apex, wherein the point or apex of the fracture points medially or laterally.
2. **Varus deformity:** The distal fragment ends are angled toward the midline of the body and the apex is pointed away from the midline.
3. **Valgus deformity:** The distal fragment ends are angled away from the midline and the apex is pointed toward the midline.

NOTE: The terms *varus* and *valgus* also are used as inversion and eversion stress movement terms (see Terminology, Chapter 1).

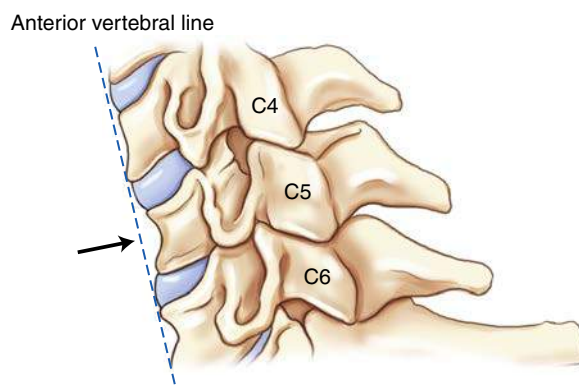


Fig. 15.4 Subluxation of cervical vertebra (C5 vertebra displaced posteriorly).

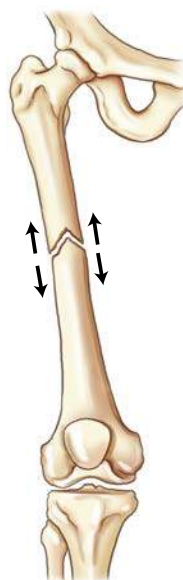


Fig. 15.5 Lack of apposition.



Fig. 15.6 Bayonet apposition.

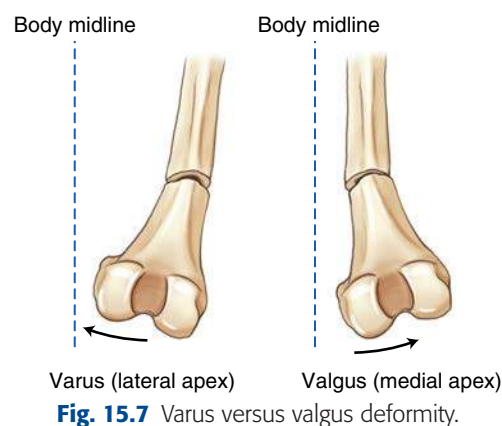


Fig. 15.7 Varus versus valgus deformity.

TYPES OF FRACTURES

Many terms are used in describing fractures. Terms that technologists are most likely to encounter are as follows.

Simple (Closed) Fracture

This is a fracture in which the bone does not break through the skin.

Compound (Open) Fracture

This is a fracture in which a portion of the bone (usually the fragmented end) protrudes through the skin (Fig. 15.8).

Incomplete (Partial) Fracture

This fracture does not traverse through the entire bone. (The bone is not broken into two pieces.) It is most common in children. Two major types of incomplete fractures are as follows:

1. **Torus fx (buckle fx):** This buckle of the cortex (outer portion of the bone) is characterized by localized expansion or torus of the cortex, possibly with little or no displacement, and no complete break in the cortex.
2. **Greenstick fx (hickory or willow stick fx):** Fracture is on one side only. The cortex on one side of the bone is broken and the other side is bent. When the bone straightens, a faint fracture line in the cortex may be seen on one side of the bone, and a slight bulging or wrinkle-like defect is seen on the opposite side (Fig. 15.9).

Complete Fracture

In this fracture, the break is complete and includes the cross-section of bone. The bone is broken into two pieces. There are three major types of complete fractures.

1. **Transverse fx:** Fracture is transverse at a near right angle to the long axis of the bone.
2. **Oblique fx:** The fracture passes through bone at an oblique angle.
3. **Spiral fx:** In this fracture, the bone has been twisted apart and the fracture spirals around the long axis (Fig. 15.10).

Comminuted Fracture

In this fracture, the bone is splintered or crushed at the site of impact, resulting in two or more fragments (Fig. 15.11). Three types of comminuted (*kom-i-nu-ted*) fractures have specific implications for treatment and prognosis because of the possible substantial disruption of blood flow:

1. **Segmental fx:** A type of double fracture in which two fracture lines isolate a distinct segment of bone; the bone is broken into three pieces, with the middle fragment fractured at both ends.
2. **Butterfly fx:** A comminuted fracture with two fragments on each side of a main, wedge-shaped separate fragment; it has some resemblance to the wings of a butterfly.
3. **Splintered fx:** A comminuted fracture in which the bone is splintered into thin sharp fragments.

Impacted Fracture

In this fracture, one fragment is firmly driven into the other, such as the shaft of the bone being driven into the head or end segment. These most commonly occur at distal or proximal ends of the femur, humerus, or radius (Fig. 15.12).

SPECIFIC NAMED FRACTURES

Following are some examples and descriptions of named fractures, usually named by the type of injury or after the person who identified them.

Barton Fracture

This is an intra-articular fracture of the distal radius often associated with dislocation or subluxation of the radiocarpal joint.



Fig. 15.8 Compound fx (tibia-fibula).



Fig. 15.9 Greenstick fx (ulna).



Fig. 15.10 Spiral fx (femur).



Fig. 15.11 Comminuted fx (tibia).



Fig. 15.12 Impacted fx (radius).

Baseball (Mallet) Fracture

This fracture of the distal phalanx is caused by a ball striking the end of an extended finger. The distal interphalangeal (DIP) joint is partially flexed, and an avulsion fracture is frequently present at the posterior base of the distal phalanx.

Bennett Fracture

This longitudinal fracture, which occurs at the base of the first metacarpal with the fracture line entering the carpometacarpal joint, generally includes a posterior dislocation or subluxation.

Boxer Fracture

This fracture usually involves the distal fifth metacarpal, with an apex posterior angulation best demonstrated on the lateral view. It results from punching someone or something.

Colles Fracture

This fracture of the wrist, in which the distal radius is fractured with the distal fragment displaced posteriorly, may result from a forward fall on an outstretched arm (Fig. 15.13).

Smith (Reverse Colles) Fracture

This is a fracture of the wrist with the distal fragment of the radius displaced anteriorly rather than posteriorly, as in a Colles fracture. It commonly results from a backward fall on an outstretched arm (Fig. 15.14).

Hangman Fracture

This fracture occurs through the pedicles of the axis (C2), with or without displacement of C2 or C3.

Hutchinson (Chauffeur) Fracture

This is an intra-articular fracture of the radial styloid process. (The name originates from the time when hand-cranked cars would backfire, with the crank striking the lateral side of the distal forearm.)

Monteggia (mon-tej'-ah) Fracture

This fracture of the proximal half of the ulna, along with dislocation of the radial head, may result from defending against blows with the raised forearm (Fig. 15.15).

Pott's Fracture

This term is used to describe a complete fracture of the distal fibula with major injury to the ankle joint, including ligament damage and frequent fracture of the distal tibia or medial malleolus (Fig. 15.16).

ADDITIONAL FRACTURE TYPES**Avulsion Fracture**

This fracture results from severe stress to a tendon or ligament in a joint region. A fragment of bone is separated or pulled away by the attached tendon or ligament.

Blowout and/or Tripod Fracture

These fractures, which result from a direct blow to the orbit and/or maxilla and zygoma, create fractures to the orbital floor and lateral orbital margins.

Chip Fracture

This fracture involves an isolated bone fragment; however, this is not the same as an avulsion fracture because this fracture is not caused by tendon or ligament stress.

Compression Fracture

This vertebral fracture is caused by compression-type injury. The vertebral body collapses or is compressed. Generally, it is most evident radiographically by a decreased vertical dimension of the anterior vertebral body (Fig. 15.17).



Fig. 15.13 Colles fx (radius).



Fig. 15.14 Smith fx (reverse Colles fx).

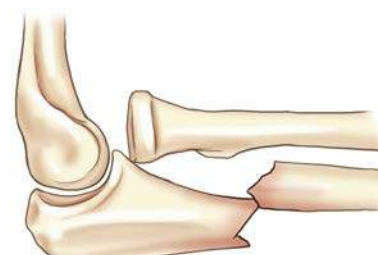


Fig. 15.15 Monteggia fx (ulna).



Fig. 15.16 Pott's fx (distal tibia-fibula).



Fig. 15.17 Compression fx (body of vertebra).

Depressed Fracture (Sometimes Called a Ping-Pong Fracture)

In this fracture of the skull, a fragment is depressed. The appearance is similar to a Ping-Pong ball that has been pressed in by the finger, but if the indentation can be elevated again, it can assume its near-original position.

Epiphyseal Fracture

This is a fracture through the epiphyseal plate, the point of union of the epiphysis and shaft of a bone. It is one of the most easily fractured sites in long bones of children. Radiologists commonly use the Salter-Harris classification (Salter 1 to 5, with Salter 5 indicating the most complex) to describe the severity and reasonable indication of prognosis of these fractures.

Pathologic Fracture

These fractures are due to disease process within the bone, such as osteoporosis, neoplasia, or other bone diseases.

Stellate Fracture

In this fracture, the fracture lines radiate from a central point of injury with a starlike pattern. The most common example of this type of fracture occurs at the patella and is often caused by knees hitting the dashboard in a motor vehicle accident (Fig. 15.18).

Stress or Fatigue Fracture (Sometimes Called a “March” Fracture)

This type of fracture is nontraumatic in origin. It results from repeated stress on a bone, such as from marching or running. If caused by marching, these fractures usually occur in the midshafts of metatarsals; if caused by running, they are in the distal shaft of the tibia. Stress fractures are frequently difficult to demonstrate radiographically and may be visible only through subsequent callus formation at the fracture site or on a nuclear medicine bone scan.

Trimalleolar Fracture

This fracture of the ankle joint involves the medial and lateral malleoli as well as the posterior lip of the distal tibia.

Tuft or Burst Fracture

This comminuted fracture of the distal phalanx may be caused by a crushing blow to the distal finger or thumb (Fig. 15.19).

POSTFRACTURE REDUCTION**Closed Reduction**

Fracture fragments are realigned by manipulation and are immobilized by a cast or splint. A closed reduction is a nonsurgical procedure; however, it may be done with the aid of fluoroscopy.

Open Reduction

For severe fractures with significant displacement or fragmentation, a surgical procedure is required. The fracture site is exposed and screws, plates, or rods are installed as needed to maintain alignment of the bony fragments until new bone growth can take place. This is called an open reduction with internal fixation (ORIF), as described later (see the section Surgical Radiography).

Mobile X-Ray Equipment

A study of trauma and mobile radiography requires an understanding of the functions and operations of the equipment being used. Trauma radiography may be performed with a conventional overhead tube in a dedicated trauma bay located in the ED or with **mobile (portable) units** that are brought to the ED, the patient's bedside, or the operating room (OR) for surgical procedures (Fig. 15.20). Radiographic examinations may also be performed in the

radiology department if the physician has deemed that the patient is stable. In this scenario, a nurse may accompany the patient to monitor the person's condition.

MOBILE X-RAY SYSTEMS

Major advances have been made in mobile radiographic and fluoroscopic equipment. Examples of general types commonly used are described and illustrated in this chapter.

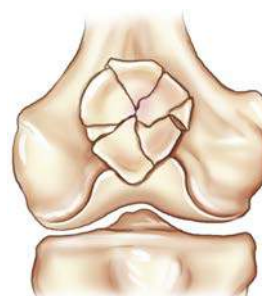


Fig. 15.18 Stellate fx (patella).



Fig. 15.19 Tuft fx (distal phalanx).



Fig. 15.20 GE Image Optima XR220amx. (Courtesy GE Healthcare.)

Battery-Driven, Battery-Operated, Mobile X-Ray Units

These systems are powered by 10 to 16 rechargeable, sealed, lead acid-type 12-volt batteries connected in series. The self-propelled systems of these units are also battery powered and have variable travel speeds up to an average walking speed of 2½ to 3 mph with a maximum incline of 7°. They have a driving range of up to 10 miles on a level surface after a full charge.

These units are driven and maneuvered by dual-drive motors that operate the two drive wheels. They also have a lower speed forward and reverse for maneuvering in close quarters. Parking brakes are automatically engaged when the control levers are not in use; this is known as dead man control. If the technologist releases the control levers, the mobile unit will come to an abrupt halt.

The unit can be plugged in for recharging when it is not being used and can be recharged at 110 or 220 V. The parking brakes are also used during charging. With 110-V, 5-amp outlets, the charging time is about 8 hours when fully discharged (Fig. 15.21).

Standard Power Source, Capacitor-Discharge, Non-Motor-Driven Units

A second type of mobile x-ray unit without battery power is now available. These models are much lighter in weight and usually are not motor driven. They operate with a 110-V, 15-amp power source or a 220-V, 10-amp power source. These units generally incorporate a capacitor discharge system, which stores electrical charges when plugged in and then discharges this electrical energy across the x-ray tube when exposure is initiated. This increases the electrical power (voltage) from the standard 110- or 220-V power source.

Other systems offer a dual power source with both battery power and plug-in electrical power for increased output. These generally also have a battery-assisted motor drive for easier transporting (Fig. 15.22).

The controls on these units may include some type of optional programmed memory system that is based on anatomic parts, or they may have operator-selected kVp and mAs technique controls.

NOTE: These are only two examples of available mobile systems. Other manufacturers offer various modifications, features, and options.

Digital Imaging Considerations

The widespread use of digital imaging in mobile systems is the most notable advancement in mobile equipment. As discussed in Chapter 1, digital imaging includes both computed radiography (CR) and digital radiography (DR). Digital imaging is especially well suited for trauma and mobile imaging in the ED, in the OR, and for bedside (mobile/portable) examinations. These procedures frequently are performed under difficult but urgent conditions in which opportunities for repeats are limited. The wide exposure latitude of digital images has improved the consistency of these images and has greatly reduced the need for repeat exposures due to positioning and technical variables.

Another advantage of digital imaging for trauma and mobile radiographic examinations is the ability to transfer these images electronically to more than one location simultaneously for interpretation or consulting. Radiologists can view the images and arrive at a diagnosis in a very short time and can communicate those findings to the ED physician, who then is able to create a plan of care for the trauma patient. In some instances, the images can be viewed directly on the mobile unit.

Following is a summary of guidelines that should be followed when digital imaging technology (computed radiography or digital radiography) is used for the lower limbs:

1. **Four-sided collimation:** Collimate to the area of interest with a minimum of two collimation parallel borders clearly demonstrated in the image. Four-sided collimation is always preferred if the study permits.

2. **Accurate centering:** It is important that the body part and the central ray be centered to the IR.
3. **Exposure factors:** With regard to exposure to the patient, it is important that the ALARA principle (as low as reasonably achievable) be followed and the lowest exposure factors required to obtain a diagnostic image be used. This includes the highest kVp and the lowest mAs that would result in desirable image quality. It may be necessary to increase kVp over that used for analog (film-screen) imaging for larger body parts, with 50 kVp as the minimum used on any procedure (exception is mammography).
4. **Post-processing evaluation of exposure indicator:** The exposure indicator value on the final processed image must be checked to verify that the exposure factors used were in the correct range to ensure optimal quality with the least radiation to the patient. If the index is outside of the acceptable range, the technologist must adjust kVp or mAs or both accordingly for any repeat exposures.



Fig. 15.21 Carestream DRX Revolution Mobile X-ray System. (Courtesy Carestream Health.)



Fig. 15.22 Siemens Mobilett Plus—dual power source, battery and/or standard power, capacity discharge. (Courtesy Siemens Medical Solutions, Malvern, PA.)

Positioning Principles for Trauma and Mobile Radiography

Positioning principles for trauma and mobile radiography are similar to those applied for routine general radiography, as described in Chapter 1 of this text. The primary difference can be summarized by the word **adaptation**. Each trauma patient and situation is unique, and the technologist must evaluate the patient and adapt CR angles and IR placement as needed. However, all images must be as true to those of routine general radiography as possible.

The technologist must keep the following three principles in mind when performing trauma or mobile radiography:

1. Obtain two projections 90° to each other with true CR/part IR alignment
2. Include the entire structure or trauma area on the IR
3. Maintain the safety of the patient, health care workers, and the public

PRINCIPLE 1: TWO PROJECTIONS 90° TO EACH OTHER WITH TRUE CR/PART IR ALIGNMENT

Trauma radiography generally requires orthogonal views, **two projections taken at 90° (or right angles to each other)** while true CR/part IR alignment is maintained. The preferences for the two projections are a true AP or PA and a true lateral achieved by turning the body part (standard positioning) or angling the CR and IR as needed (trauma adaptation positioning). In this way, the CR/part IR alignment can be maintained even if the patient cannot be turned or rotated. An example is shown in Figs. 15.23 and 15.24, in which true AP and lateral foot images are obtained without flexing or moving the lower limb. The AP projection is achieved by angling the CR and IR in relation to the foot, thus maintaining a true CR/part IR alignment.

When adaptations are made during the performance of any radiographic image, it is important to include as much information as possible as to how the image was achieved. This information includes CR angle, projection of the beam (AP, PA, lateral, oblique, horizontal beam lateral), and upright, semiupright, or supine position.

Exception to True Anteroposterior (AP) and Lateral Principle

Because of the patient's condition, occasionally it may not be possible to maintain this standard CR/part IR relationship for true anteroposterior (AP) and lateral projections. This may be due to unavoidable obstructions such as large splints, back supports, traction bars, or other apparatus. In this case, the technologist should still attempt two projections as near 90° to each other as possible, even if the anatomic part is partially rotated. **Only as a last resort should just one projection be taken.** When these exceptions are unavoidable, a note that explains the reason for this variance in routine should be left in the patient's record and/or examination requisition.

Exception to CR/Part IR Alignment

Generally, this principle involves placing the IR at right angles or perpendicular to the CR for minimal part distortion. However, in situations such as that shown in Fig. 15.25, the CR-part relationship can be maintained, but not the part-IR relationship. In this example, the AP axial oblique cervical spine is obtained with the patient supine and the IR flat on the table under the patient. This will result in some part distortion but, in trauma radiography, it may be an acceptable option.



Fig. 15.23 AP foot (trauma adaptation positioning).



Fig. 15.24 Lateral foot.



Fig. 15.25 Trauma AP axial oblique C-spine exception. The IR is not perpendicular to the CR.

PRINCIPLE 2: ENTIRE STRUCTURE OR TRAUMA AREA INCLUDED ON IMAGE RECEPTOR

Trauma radiography mandates that the **entire structure being examined should be included on the radiographic image** to ensure that no pathology is missed. This requires selection of sufficiently large IRs or the use of more than one IR if needed.

If an examination request on a trauma patient includes the long bones of the upper or lower limbs, **both joints should be included** for possible secondary fractures away from the primary injury. An example is a post-trauma examination request for a leg (tibia-fibula) with injury to the distal region. This may require a second, smaller IR of the knee to include the proximal tibia-fibula region if the patient's leg is too long to be included on a single image. Fractures of the distal tibia also may involve a secondary fracture of the proximal fibula. This principle of including both joints is true for AP and lateral projections.

For all upper and lower limb follow-up examinations, **always include a minimum of one joint nearest the site of injury**. Few if any exceptions to this rule exist, even if the obvious fracture shown on previous images is in the midshaft region. The joint nearest the fracture site should always be included (Figs. 15.26 and 15.27).

The principle of including the entire structure, or trauma region, also applies to these larger body areas. For example, the abdomen on a large patient may require two IRs placed landscape to include the entire abdomen. This may also be true for the chest or bony thorax.

Trauma patients often arrive in a supine position, and horizontal beam projections are commonly required for the lateral projections. Care must be taken to ensure that the divergent x-ray beam does not project the body part off the IR, especially when the IR is placed on edge directly beside the patient. This is true for the spine, skull, and other parts that rest directly on the tabletop. Examples of a horizontal beam, lateral skull projection, with and without a possible spine injury, are shown in Figs. 15.28 and 15.29. **With a questionable spinal injury, the head and neck cannot be moved or elevated.** Therefore, no support or pad can be placed between the head and tabletop. If the IR is placed on edge next to the patient's head, the divergent x-ray beam will project the posterior part of the skull off the IR.

To avoid cutoff of the posterior skull in this example, the patient can be moved to the edge of the table or cart and the IR placed below the level of the tabletop (see Fig. 15.29). This may result in an increase in the object-image receptor distance (OID), with resultant magnification. In these cases, this is an acceptable option. If cervical spine radiographs have ruled out cervical fracture or subluxation, the head may be raised and supported by a sponge to prevent posterior skull cutoff (see Fig. 15.28).



Fig. 15.26 AP distal lower leg and ankle.



Fig. 15.27 Lateral distal lower leg and ankle.

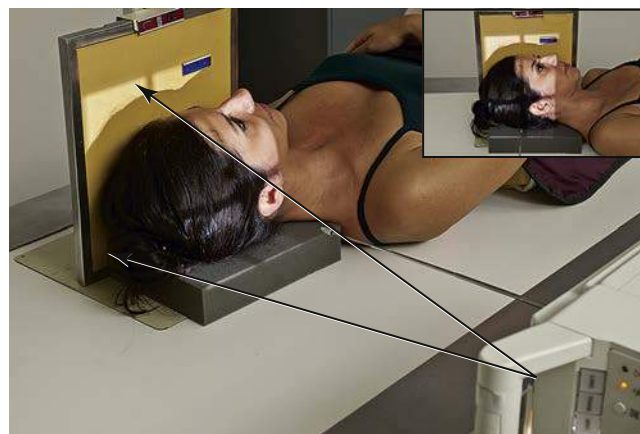


Fig. 15.28 Horizontal beam lateral skull without possible spine injury (head raised from tabletop).

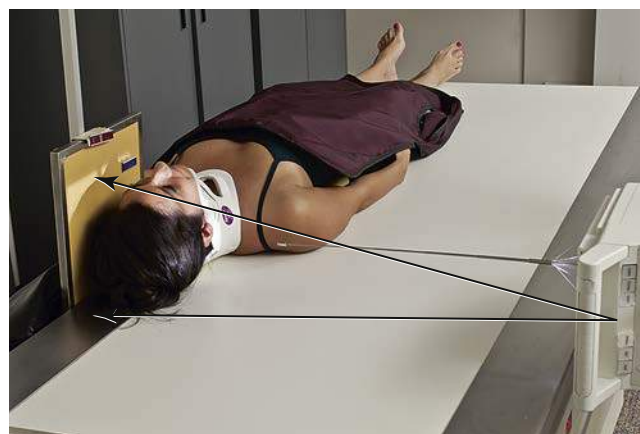


Fig. 15.29 With possible spine injury, head cannot be raised or moved (IR is placed below tabletop level to prevent posterior skull cutoff).

PRINCIPLE 3: MAINTAIN THE SAFETY OF THE PATIENT, HEALTH CARE WORKERS, AND THE PUBLIC

When trauma or mobile radiography is performed, it may be necessary to move room equipment and side rails to provide access to the patient. The technologist must take note of the equipment that is moved to make sure that it is not attached to the patient. Often, the technologist will be able to move equipment only a short distance because of space constraints. Side rails may have to be lowered to allow the technologist to place an IR under the patient. This must take place as quickly and as safely as possible. Never assume that a patient is unable to move. All side rails must be returned to the upright position, regardless of whether or not the patient already had the rails up. All equipment must be returned to its original location as well.

When performing mobile studies, **technologists are also responsible for ensuring the safety of the other health care workers in the immediate area.** In a trauma situation, time is of the essence. Although it is important that the technologist obtain trauma images while physicians, nurses, and other staff are attending to the patient, under no circumstances should an exposure take place with an unshielded person in the vicinity of the primary beam. The ALARA principle applies to other health care workers and the public, in addition to the patient.

Alternative Modalities**COMPUTED TOMOGRAPHY (CT)**

The increased speed of computed tomography (CT) scanners has contributed to their increased use for emergency imaging. CT is commonly used for accurate diagnosis of a wide range of traumatic conditions that affect all body systems, thus replacing some of the traditionally ordered diagnostic examinations, such as spine and skull radiography. The three-dimensional reconstruction capability of CT is useful for fully assessing skeletal trauma.

SONOGRAPHY (MEDICAL ULTRASOUND)

Sonography (medical ultrasound) is indicated in the early assessment of certain trauma patients, such as those who have experienced blunt abdominal injury. It is a noninvasive technique used to detect free fluid or blood in the abdomen. Sonography is also the modality of choice for imaging emergency conditions of the female reproductive system (e.g., ectopic pregnancy). Sonography is used as required for specific emergency situations when other abdominal organs are imaged.

NUCLEAR MEDICINE (NM)

NM is useful for the evaluation of specific emergency conditions, such as pulmonary embolus, testicular torsion, and gastrointestinal (GI) bleeding. Blood flow to the areas under investigation is assessed through injection of the radionuclide.

ANGIOGRAPHY AND INTERVENTIONAL PROCEDURES

Angiography is indicated for studies of the aortic arch in the trauma patient, although the use of these procedures has declined because of the increased use of CT angiography. Interventional procedures performed on the trauma patient, as described in [Chapter 17](#), include transcatheter embolization to occlude hemorrhaging vessels.

Routine and Special Projections

Certain routine and special projections for trauma, mobile, and surgical radiography are demonstrated and described on the following pages as suggested standard routine and special departmental routines or procedures.

TRAUMA AND MOBILE POSITIONING

AP CHEST

WARNING: With possible spinal injury or severe trauma, do not attempt to move the patient.

Clinical Indications

- According to the American College of Radiology (ACR), the AP chest radiograph is a standard part of the trauma workup at most level I trauma centers across the United States.³
- Approximately 25% of deaths from blunt trauma arise from chest injuries, although up to 50% of deaths are at least partially related to thoracic injuries.²
- Chest injuries include acute aortic injury, pulmonary injury, pneumothorax, hemothorax, extrapleural hematoma, large airway rupture, hemidiaphragmatic rupture, and musculoskeletal injury.³
- Ensure proper placement of lines and tubes.

Chest
TRAUMA
• AP
OPTIONAL
• Lateral
• Lateral decubitus (AP)

Technical Factors

- Minimum SID—40 (100 cm). Use 72 inches (180 cm) if possible.
- IR size—14 × 17 inches (35 × 43 cm), **landscape**, for average to large patients (see NOTE 1)
- Grid (see NOTE 2)
- kVp range—90–125, depending on whether grid is required

Patient Position

AP Chest

- Patient is supine on cart; if patient's condition allows, the head end of the cart should be raised into an erect or semierect position.
- Rotate arms internally, if patient's condition allows, to move scapulae out of lung fields.

Part Position

AP Chest

- Enclose the IR in a plastic cover and place under or behind patient; place top of IR about 2 inches (4 to 5 cm) above the shoulders, aligning CR to IR. Ensure **no rotation** (coronal plane parallel to IR). (Place supports under parts of IR as needed.)

CR

AP Chest

- Direct CR 3 to 4 inches (8 to 10 cm) below jugular notch, **level of T7**.
- Angle CR 3° to 5° caudad, or raise head end of bed slightly, to place the CR perpendicular to long axis of sternum (unless grid prevents this). This simulates the PA projection and prevents the clavicles from obscuring the apices of the lungs (Fig. 15.30).
- If patient is able to attain only a semierect position, the CR must be angled to maintain the perpendicular relationship with the IR (Fig. 15.31).

Radiation Safety

- Exposure factor selection should be optimized in accordance with ALARA.
- Collimate on four sides to anatomy of interest.
- Shield radiosensitive tissues outside area of interest.

Respiration Expose at end of second full inspiration.

Optional lateral chest (not demonstrated here): A lateral image can be obtained with a horizontal beam CR if patient can raise arms at least 90° from body. Place IR parallel to midsagittal plane (MSP), with top of IR 2 inches (5 cm) above level of shoulders. Support patient on radiolucent pad to center chest to IR and center horizontal CR to level of T7.

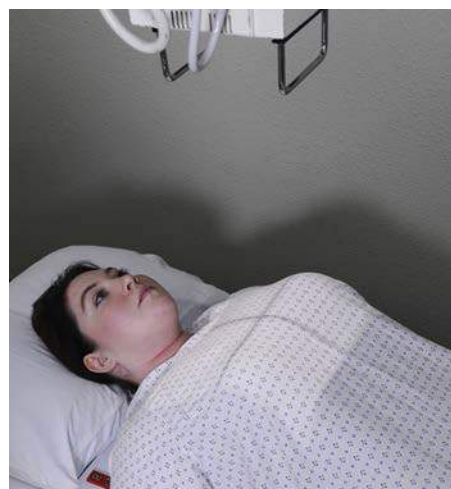


Fig. 15.30 AP supine chest, bedside (IR landscape), CR 3° to 5° caudad, perpendicular to sternum.

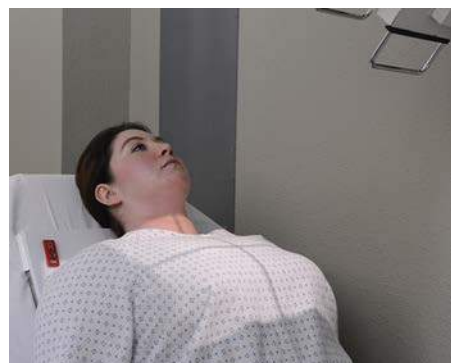


Fig. 15.31 AP semierect chest, bedside.

Lateral Decubitus AP Projection To determine air-fluid levels when the patient cannot be elevated sufficiently for an erect position, a lateral decubitus can be taken in bed with the IR placed behind the patient or on a stretcher in front of the IR holder, as shown in Fig. 15.32. Place radiolucent pads under the thorax and shoulders, and raise the arms above the head. CR/part IR alignment and centering are similar to that for supine AP, with necessary adaptations for the decubitus position.

NOTE 1: Costophrenic angle cutoff is a problem with recumbent chest positions taken with a shorter source–image receptor distance (SID) because of the divergence of the x-ray beam. Therefore, unless the patient is quite small, a **landscape IR placement is recommended**.

NOTE 2: Focused grids may be difficult to use for mobile chests because of the problems of grid cutoff.



Fig. 15.32 Lateral decubitus (AP) chest, horizontal beam for detecting possible air-fluid levels.

AP SUPINE AND DECUBITUS: ABDOMEN

WARNING: With possible spinal injury or severe trauma, do not attempt to move the patient.

Clinical Indications

- Evaluate for fracture, free intraperitoneal air, abnormal fluid or gas.⁴
- Chest radiograph may be ordered along with an abdominal series.

Abdomen
TRAUMA

- AP supine
- Decubitus

Technical Factors

- Minimum SID—40 inches (100 cm)
- IR size—14 × 17 inches (35 × 43 cm), portrait (see NOTE 1)
- Grid
- kVp range—70–90
- Include decubitus and upside markers if applicable

Patient Position

AP Supine (Fig. 15.33)

- Place IR into plastic cover if taken at bedside.
- Align IR portrait to MSP.
- Arms placed at side, away from the body.

Left Lateral Decubitus AP (or PA) Projection (Fig. 15.34)

- This projection allows determination of air-fluid levels (see NOTE 2) and possible free intra-abdominal air when an upright image is not possible. The lateral decubitus can be taken in bed, on a stretcher in the ED, or on a stretcher in the radiography room in front of an upright wall bucky.
- Place supports or a positioning board under hips and thorax as needed to center abdomen to IR for lateral and dorsal decubitus projections, if performed bedside. This will create a flat surface, thus preventing the patient from sinking into the mattress and cutting off downside anatomy on the image.
- Raise arms up near head and partially flex the knees to stabilize patient.

Part Position

AP Supine

- Center IR to CR at level of iliac crest. Ensure that both sides of upper and lower abdomen are at equal distances from lateral IR margins.
- Place supports under parts of IR if needed to ensure that IR is level and perpendicular to CR (prevents patient rotation and grid cutoff on soft bed surfaces).

Left Lateral Decubitus AP (or PA) Projection

- Ensure that **diaphragm and upper abdomen are included**. Place center of IR 1 to 2 inches (2.5 to 5 cm) above level of iliac crests.
- Ensure no rotation and that the IR plane is perpendicular to CR.

CR

AP Supine

- Position CR perpendicular to **level of iliac crest** and to center of IR.

Left Lateral Decubitus AP (or PA) Projection

- Direct horizontal CR to center of IR, 2 inches (5 cm) above level of iliac crests.

Radiation Safety

- Exposure factor selection should be optimized in accordance with the ALARA.
- Collimate on four sides to anatomy of interest.
- Shield radiosensitive tissues outside area of interest.



Fig. 15.33 Supine AP abdomen, bedside.



Fig. 15.34 Left lateral decubitus (AP) abdomen, bedside.

Respiration**AP Supine and Left Lateral Decubitus AP (or PA) Projection**

- Make exposure at end of expiration.

NOTE 1: For patients with a large body habitus, two 14 × 17-inch (35 × 43-cm) IRs may be needed to ensure the entire anatomy is included. These IRs should be placed landscape, one for imaging the upper abdomen and diaphragm and the other for imaging the lower abdomen and symphysis pubis. Two separate exposures are necessary (see [Chapter 3](#)).

NOTE 2: For lateral decubitus projections, have patients lie on the side for a minimum of **5 minutes** before taking exposure to allow air to rise to highest position within the abdomen.

Dorsal Decubitus, Lateral Position (Fig. 15.35)

- This is not a common bedside projection. The dorsal decubitus is a useful position for demonstrating a possible abdominal aortic aneurysm, or it may be used as an alternative to the lateral decubitus position if the patient cannot be moved.

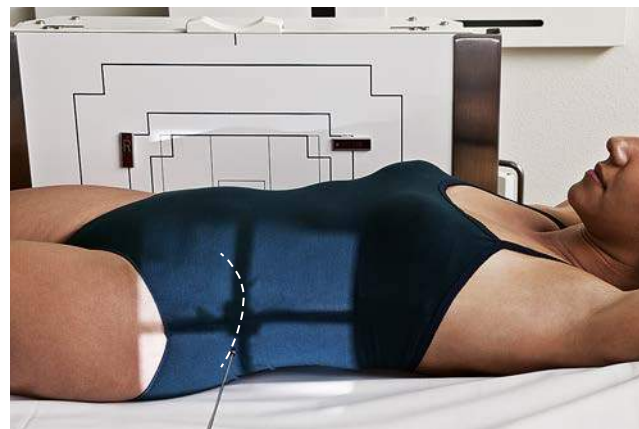


Fig. 15.35 Dorsal decubitus (lateral) abdomen, on stretcher in front of erect bucky.

UPPER LIMB

WARNING: With possible spinal injury or severe trauma, do not attempt to move the patient.

Clinical Indications

- Fractures, dislocations, and subluxations due to trauma.
- Hand and wrist fractures and dislocations are more common than those of any other part of the body.⁵
- Delayed diagnosis due to persistent clinical findings.

Upper Limb

TRAUMA

- AP or PA
- Lateral

OPTIONAL

- Oblique

Technical Factors

- Minimum SID—40 inches (100 cm)
- IR size—Smallest IR possible
- Grid—If part is thicker than 10 cm
- kVp range—50–70

Patient Position

- The patient will present in a multitude of positions. Some may be able to sit upright for their exam, whereas others may be supine on a back board. Because of the wide variance in patient positions, it is difficult to describe a specific protocol to follow in regard to patient positioning. The technologist must assess the patient's status and limitations to determine how to proceed in the best interest of the patient.

Part Position

- Obtain a minimum of two projections 90° to each other with true CR/part IR alignment (Figs. 15.36 and 15.37).
- Do not attempt to rotate a severely injured part.
- Do not remove splints or other immobilization devices.
- Be cautious working around foreign bodies that may be protruding from the area of interest.
- When placing the IR, minimally raise the affected limb while supporting both joints.

CR

- Include the entire structure or trauma area, including both joints for long bones.

Radiation Safety

- Exposure factor selection should be optimized in accordance with the ALARA.
- Collimate on four sides to anatomy of interest.
- Shield radiosensitive tissues outside area of interest.



Fig. 15.36 PA forearm to include wrist and elbow.

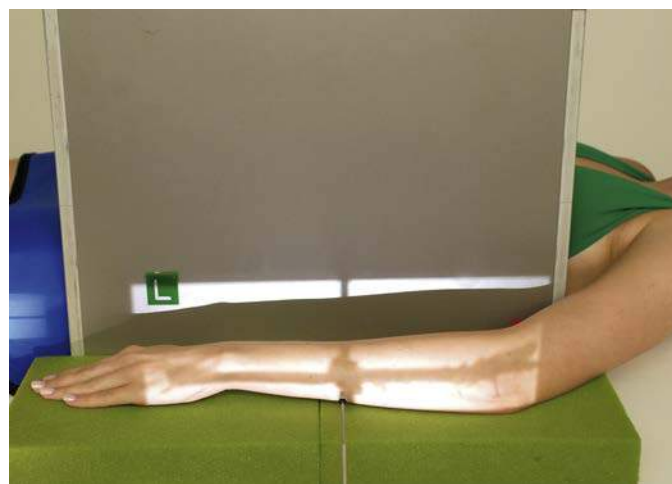


Fig. 15.37 Lateral forearm to include wrist and elbow.

Upper Limb Positioning Examples Figs. 15.38 to 15.50 provide examples of how radiographs can be produced when encountering a trauma situation involving the upper limb.



Fig. 15.38 AP proximal metacarpals and wrist.



Fig. 15.42 Lateral wrist and hand.



Fig. 15.39 AP hand and fingers for distal phalanges.



Fig. 15.43 PA thumb.



Fig. 15.40 AP hand.

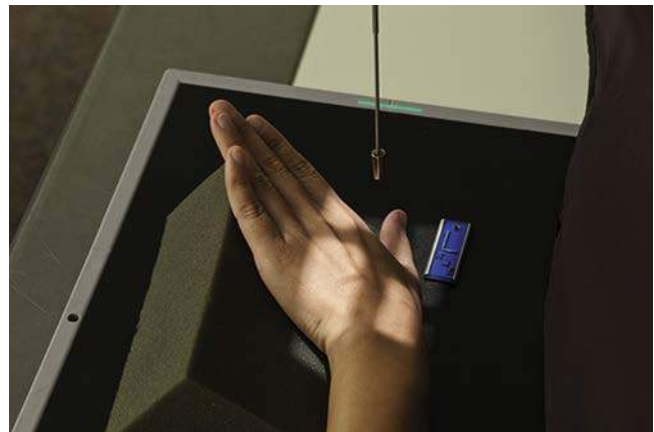


Fig. 15.44 Lateral thumb.



Fig. 15.41 Oblique—fingers, hand, and/or wrist.



Fig. 15.45 PA horizontal beam elbow, CR perpendicular to interepicondylar plane.



Fig. 15.48 Lateral for coronoid process, elbow flexed 80°, CR angled 45° distally (from shoulder).



Fig. 15.46 Lateral elbow partially flexed, CR angled as needed to be parallel to interepicondylar plane.



Fig. 15.49 AP humerus; should include both joints.



Fig. 15.47 Lateral for radial head, elbow flexed 90°, CR angled 45° proximally (toward shoulder).

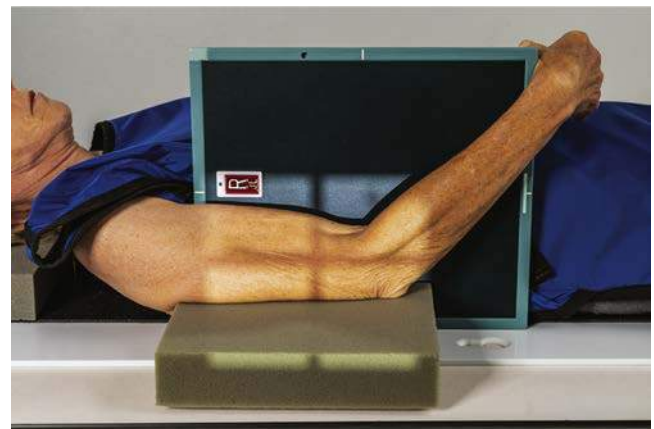


Fig. 15.50 Lateral, mid- and distal humerus to include elbow.

SHOULDER

WARNING: Do not attempt to rotate the arm if a fracture or dislocation is suspected; leave affected arm as presented.

NOTE: Local protocols for radiographic evaluation of the shoulder for trauma vary widely. However, the shoulder trauma protocol should have at least three views, two of which are orthogonal.⁶

Shoulder

TRAUMA

- AP
- Scapular Y lateral
- Inferosuperior axial
- Transthoracic lateral

Clinical Indications

- Fractures, dislocations, and subluxations due to trauma

Technical Factors

- Minimum SID—40 inches (100 cm)
- IR size—10 × 12 inches (24 × 30 cm) portrait
- Grid
- kVp range—70–85 with grid

Patient Position

AP Shoulder and Scapular Y Lateral—AP Oblique (Lateromedial Scapula) (Figs. 15.51 and 15.52)

- Patient will most likely be supine in the case of trauma; however, an erect position is usually more comfortable. Cervical spine and lower limb injuries must be ruled out before asking patient to stand or sit up.

Inferosuperior Axial (Fig. 15.53)

- Patient supine, with shoulder raised approximately 2 inches (5 cm) from the cart or tabletop.
- Place support under arm and shoulder to place area of interest near center of IR.

Transthoracic Lateral (Fig. 15.54)

- Patient will most likely be supine in case of trauma; however, an erect position is usually more comfortable.
- Place the affected shoulder closest to the IR.

Part Position

AP Shoulder

- Affected arm in neutral rotation, position at side.
- Center IR (grid IR under patient if on stretcher) to shoulder joint and to CR.

Scapular Y Lateral—AP Oblique (Lateromedial Scapula)

- Palpate borders of scapula by grasping medial and lateral borders of body of scapula with fingers and thumb. Carefully adjust body rotation as needed to bring the **plane of the scapular body perpendicular to the IR (approximately 25° to 30° away from IR)**. Do not roll patient until the cervical spine has been cleared for fracture or subluxation.
- Center scapulohumeral joint to CR and center of IR.

Inferosuperior Axial

- Place IR as close to the neck as possible.
- Abduct affected arm 90° from body or as much as the patient can tolerate. Care should be taken when abducting the arm; support should be provided under arm. Do not abduct arm if fracture or dislocation is suspected.

Transthoracic Lateral

- Raise unaffected arm above head, elevating the unaffected shoulder.
- Center to level of surgical neck of affected arm to center of IR.



Fig. 15.51 AP shoulder.



Fig. 15.52 AP oblique, scapular Y, lateromedial projection of scapula.

CR**AP Shoulder**

- CR perpendicular to IR, directed to **midscapulohumeral joint**.

Scapular Y Lateral—AP Oblique (Lateromedial Scapula)

- Project CR perpendicular to IR, or if patient cannot be turned up sufficiently, angle CR as needed to be parallel to scapular blade (place grid horizontal to prevent grid cutoff) (see NOTE 1).
- Center CR to **scapulohumeral joint** (2 or 2½ inches [5 or 6 cm] below top of shoulder).

Inferosuperior Axial

- Direct CR **medially 15° to 30°** (less angle is required with less abduction of the arm).
- Center the CR **horizontally to axilla and humeral head**.

Transthoracic Lateral

- CR perpendicular to IR, directed through thorax **exiting at surgical neck** of affected arm (see NOTE 2).

Radiation Safety

- Exposure factor selection should be optimized in accordance with the ALARA.
- Collimate on four sides to anatomy of interest.
- Shield radiosensitive tissues outside area of interest.

Respiration

- Suspend respiration.
- A breathing technique may be preferred for the transthoracic lateral to blur out ribs and lung structures.

NOTE 1: Some distortion will occur with this medial CR angle if it is needed to achieve a lateral position of the scapula.

NOTE 2: A 10° to 15° cephalad angle may be required if superimposition of the shoulders occurs.



Fig. 15.53 Inferosuperior axial (transaxillary) shoulder.



Fig. 15.54 Lateral, transthoracic proximal humerus.

LOWER LIMB

WARNING: With possible spinal injury or severe trauma, do not attempt to move the patient.

NOTE: Radiographs are the mainstay of initial medical imaging in the setting of acute foot trauma. Initial foot imaging typically consists of a three-view study with the possibility of additional views as indicated by the clinical setting.⁷ An evaluation of the traumatized ankle should consist of AP, lateral, and mortise views of the ankle.⁸ Several studies have found that the knee radiograph is commonly obtained after trauma but has the lowest yield for diagnosing clinically significant fractures.⁹

Lower Limb

TRAUMA

- AP
- Lateral

OPTIONAL

- Oblique

Clinical Indications

- Fractures, dislocations, and subluxations due to trauma

Technical Factors

- Minimum SID—40 inches (100 cm)
- IR size—Smallest IR possible
- Grid—If part is thicker than 10 cm
- kVp range—60–85

Patient Position

- The patient will present in a multitude of positions. Because of the wide variance in patient positions, it is difficult to describe a specific protocol to follow in regard to patient positioning. The technologist must assess the patient's status and limitations to determine how to proceed in the best interest of the patient.

Part Position

- Obtain a minimum of two projections 90° to each other with true CR/part IR alignment (Figs. 15.55 and 15.56).
- Do not attempt to rotate a severely injured part.
- Do not remove splints or other immobilization devices.
- Be cautious working around foreign bodies that may be protruding from the area of interest.
- When placing the IR, minimally raise the affected limb while supporting both joints.

CR

- Include the entire structure or trauma area, including both joints for long bones.

Radiation Safety

- Exposure factor selection should be optimized in accordance with the ALARA.
- Collimate on four sides to anatomy of interest.
- Shield radiosensitive tissues outside area of interest.



Fig. 15.55 AP knee, CR parallel to long axis of foot lateromedially. (No cephalic angle is required on average patients.)



Fig. 15.56 Lateromedial knee, horizontal CR.

LOWER LIMB POSITIONING EXAMPLES

Figs. 15.57 to 15.67 provide examples of how radiographs can be produced when encountering a trauma situation involving the lower limb.



Fig. 15.57 AP foot and/or toes—CR perpendicular to IR.



Fig. 15.60 Optional—AP ankle, CR perpendicular (parallel to long axis of foot).



Fig. 15.58 Optional—oblique foot, CR cross-angled lateromedially 30° to 40°.

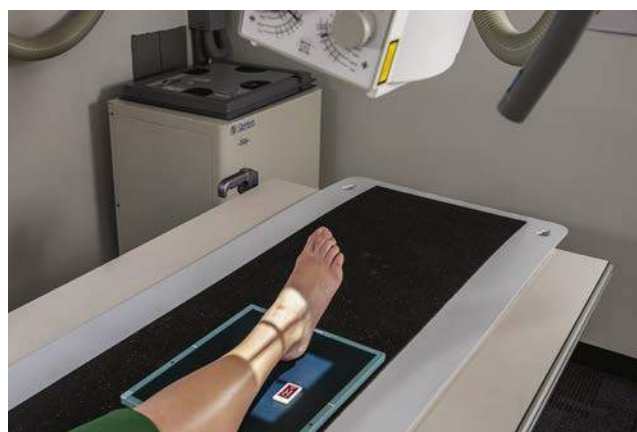


Fig. 15.61 AP mortise projection—CR 15° to 20° lateromedial angle, perpendicular to intermalleolar plane.



Fig. 15.59 Lateral foot or calcaneus.



Fig. 15.62 Lateral ankle—CR horizontal.



Fig. 15.63 AP lower leg—CR cross-angled lateromedially (parallel to long axis of foot).



Fig. 15.66 AP mid- and distal femur.



Fig. 15.64 Lateral lower leg.

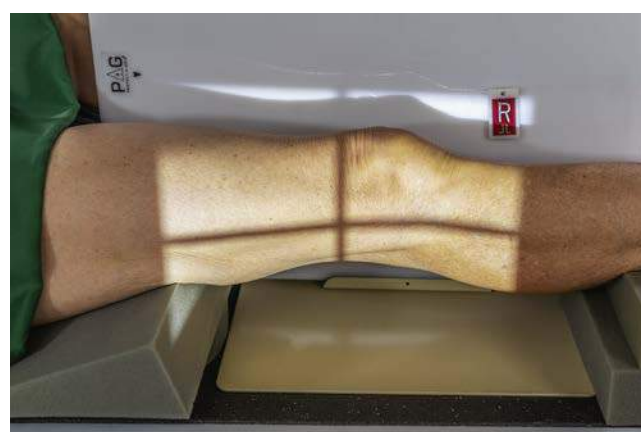


Fig. 15.67 Lateral mid- and distal femur.



Fig. 15.65 Optional medial oblique knee—CR 45° lateromedial cross-angle, grid landscape (crosswise).

PELVIS

WARNING: Do not attempt to rotate leg internally if hip fracture is suspected.

NOTE: An AP pelvic radiograph is often combined with an AP chest and a lateral horizontal beam cervical spine radiograph to quickly assess the patient for emergent injuries and to triage patients.³

Pelvis
TRAUMA
• AP

Clinical Indications

- Fractures, dislocations, and subluxations due to trauma

Technical Factors

- Minimum SID—40 inches (100 cm)
- IR size—14 × 17 inches (35 × 43 cm), landscape
- Grid
- kVp range—75–90

Patient Position

AP Pelvis

- Patient supine, with arms removed from area of interest (Fig. 15.68).
- Direct CR perpendicular to center of IR and pelvis.

Part Position



AP Pelvis

- Place plastic cover over IR and slide under pelvis, **IR landscape**, centered to patient.
- Top of IR will be about 1 inch (2.5 cm) above iliac crest.
- Ensure **no rotation** and equal distances from anterior superior iliac spine (ASIS) to IR.
- Rotate feet internally 15° if possible (see previous WARNING).

CR

AP Pelvis

- CR is perpendicular to IR, directed **midway between level of ASIS and the symphysis pubis**. This is approximately 2 inches inferior to the level of ASIS.

Radiation Safety

- Exposure factor selection should be optimized in accordance with the ALARA.
- Collimate on four sides to anatomy of interest.
- Shield radiosensitive tissues outside area of interest.

Respiration Suspend respiration during exposure.



Fig. 15.68 AP pelvis—bedside mobile. (Right leg is not rotated internally in this example.)

HIP

WARNING: Do not attempt to rotate leg internally if hip fracture is suspected.

NOTE: Radiography is the established initial imaging study of choice for assessing the acutely painful hip. As with any trauma-related musculoskeletal radiographic studies, orthogonal projections (two or more views of the anatomy at right angles to each other) are considered standard.¹⁰

Hip

TRAUMA

- AP
- Lateral-axiolateral-inferosuperior hip—Danelius-Miller method

Clinical Indications

- Fractures, dislocations, and subluxations due to trauma

Technical Factors

- Minimum SID—40 inches (100 cm)
- IR size—14 × 17 inches (35 × 43 cm) or 10 × 12 inches (24 × 30 cm), portrait
- Grid
- kVp range—75–90

Patient Position**AP Hip and Axiolateral-Inferosuperior Hip—Danelius-Miller Method**

- Patient supine, with arms removed from area of interest.

Part Position **AP Hip (Fig. 15.69)**

- Place 14 × 17-inch (35 × 43-cm) portrait under hip with the top of the IR at the level of the iliac crest.
- Ensure no **rotation** and equal distances from anterior superior iliac spine (ASIS) to IR
- Rotate leg 15° internally, if possible (see previous WARNING).

Axiolateral-Inferosuperior Hip—Danelius-Miller Method (Fig. 15.70)

- Place IR against patient's side just above iliac crest and adjust so that it is **parallel to femoral neck**.
- Rotate leg 15° internally, if possible (see previous WARNING).
- Elevate opposite leg as much as possible.

CR**AP Hip**

- CR is perpendicular to IR, directed 1 to 2 inches (2.5 to 5 cm) distal to midfemoral neck.

Axiolateral-Inferosuperior Hip—Danelius-Miller Method

- Direct horizontal CR perpendicular to femoral neck and to plane of IR.
- Ensure that CR is centered to the midline of the grid to prevent cutoff.

Radiation Safety

- Exposure factor selection should be optimized in accordance with the ALARA.
- Collimate on four sides to anatomy of interest.
- Shield radiosensitive tissues outside area of interest.

Respiration Suspend respiration during exposure.



Fig. 15.69 AP hip.



Fig. 15.70 Inferosuperior lateral—bedside mobile (Danelius-Miller method).

CERVICAL SPINE

WARNING: Do not remove cervical collar or move patient's head or neck until cervical fractures have been ruled out.

NOTE: A meta-analysis of seven studies that met strict inclusion criteria revealed that the pooled sensitivity of radiography for detecting cervical spine injuries (CSIs) was 52%, whereas the combined sensitivity of CT was 98%. Screening the cervical spine with multidetector CT (MDCT) is faster than performing radiography, with far fewer technical failures. The ACR panel concluded that thin-section MDCT, and not radiography, should be the primary screening study for suspected CSI. When motion artifacts are significant enough to prevent adequate evaluation of vertebral integrity, a single lateral view will suffice to show that there is normal alignment and no evidence of fracture.¹¹

Cervical Spine

TRAUMA

- Lateral, horizontal beam

OPTIONAL

- Cervicothoracic (swimmer) lateral

Clinical Indications

- Fractures, dislocations, and subluxations due to trauma

Technical Factors

- SID—60–72 inches (150–180 cm)
- IR size—10 × 12 inches (24 × 30 cm), portrait
- Grid or without grid (see **NOTE 1**)
- Specially designed compensating filter useful for obtaining uniform brightness on the swimmer lateral (see **Chapter 1**).
- kVp range—70–85; lateral, horizontal beam; 75–95; cervicothoracic (swimmer) lateral

Patient Position

Lateral, Horizontal Beam and Cervicothoracic (Swimmer) Lateral

- Patient supine with potential spinal injury

Part Position

Lateral, Horizontal Beam (Fig. 15.71)

- Vertical IR against shoulder, parallel to MSP, with **top of IR 1 to 2 inches (2.5 to 5 cm) above level of EAM**. Ensure that C7–T1 region is included.
- Have patient relax and depress shoulders as much as possible (see **NOTE 2**).

Cervicothoracic (Swimmer) Lateral (Fig. 15.72)

- Vertical IR placement is similar to lateral, horizontal beam placement.
- Elevate arm and shoulder closest to IR and depress opposite shoulder as much as possible (see **NOTE 3**).

CR

Lateral, Horizontal Beam

- Direct CR **horizontal to C4** (upper thyroid cartilage) and to center of grid to prevent grid cutoff, or turn grid with centerline vertical to prevent grid cutoff if necessary.

Cervicothoracic (Swimmer) Lateral

- Direct CR horizontal, **centered to C7–T1** (approximately 1 inch [2.5 cm] above level of jugular notch). Center the center of grid to CR to prevent grid cutoff (grid lines vertical).

Radiation Safety

- Exposure factor selection should be optimized in accordance with the ALARA.
- Collimate on four sides to anatomy of interest.
- Shield radiosensitive tissues outside area of interest.

Respiration

Suspend respiration on full expiration.

NOTE 1: It is estimated that, for a small body part (4 inches [10 cm]), a 10-inch (25-cm) air gap will clean up scatter as well as a 15:1 grid. The cleanup is not as efficient for a larger body part (8 inches [20 cm]).¹²

NOTE 2: Traction on the arms will help depress the shoulders but should be done only by a qualified assistant and/or with the consent or assistance of a physician. If the C7–T1 junction cannot be visualized on the initial horizontal beam lateral C-spine image, a horizontal beam swimmer's lateral should be performed.

NOTE 3: A 5° CR caudal angle may be required if patient cannot depress shoulder opposite IR.



Fig. 15.71 Horizontal beam lateral, C spine.



Fig. 15.72 Swimmer's lateral—C–T1.

THORACIC AND LUMBAR SPINE

WARNING: With possible spinal injury or severe trauma, do not attempt to move the patient.

Lateral projections are typically reviewed by a physician prior to obtaining additional spine images as described in Chapters 8 and 9.

NOTE: Currently MDCT is the imaging procedure of choice for evaluating trauma patients with possible spinal fractures or injuries.¹¹

Thoracic and Lumbar Spine
TRAUMA
• Lateral, horizontal beam
OPTIONAL
• Cervicothoracic (swimmer) lateral (previous page)

Clinical Indications

- Fractures, dislocations, and subluxations due to trauma

Technical Factors

- Minimum SID—40 inches (100 cm)
- IR size—14 × 17 inches (35 × 43 cm), portrait
- Grid (see NOTE 1)
- kVp range—80–95 lateral thoracic spine; 90–95 lateral lumbar spine

Patient Position

Lateral Thoracic Spine, Horizontal Beam and Lateral Lumbar Spine, Horizontal Beam

- Patient supine, raise arms sufficiently so as to not obscure anatomy of interest.

Part Position

Lateral Thoracic Spine, Horizontal Beam and Lateral Lumbar Spine, Horizontal Beam

- Build up patient with backboard (see Fig. 15.73) or move patient to edge of table and place vertical IR below level of tabletop. Use IR holder or tape and/or sandbags to support IR (see NOTE 2).

CR

Lateral Thoracic Spine, Horizontal Beam

- Center horizontal CR to vertebral column (midway between midcoronal plane and posterior aspect of thorax); near centerline of grid at level of T7, 3 to 4 inches (8 to 10 cm) inferior to jugular notch (see Fig. 15.73).

Lateral Lumbar Spine, Horizontal Beam

- Center horizontal CR centered to vertebral column (midcoronal plane); near centerline of grid at level of L4 or the iliac crest (see Fig. 15.74).

Radiation Safety

- Exposure factor selection should be optimized in accordance with the ALARA.
- Collimate on four sides to anatomy of interest.
- Shield radiosensitive tissues outside area of interest.

Respiration

Suspend respiration on full expiration.

NOTE 1: A decubitus-type grid with lead strips aligned horizontally can be used to prevent grid cutoff. The grid then may be placed landscape to patient, for better centering of the CR to the near centerline of the grid. This applies to both horizontal beam thoracic and lumbar spine projections.

NOTE 2: Both lateral thoracic and lumbar examinations can be performed with the patient remaining on the gurney. The stretcher can be moved to an upright bucky and the CR aligned with the patient positioned appropriately.



Fig. 15.73 Horizontal beam lateral—thoracic spine.



Fig. 15.74 Horizontal beam lateral—lumbar spine.

CRANIUM—LATERAL, HORIZONTAL BEAM

WARNING: Cervical spine fractures and subluxations or dislocations must be ruled out before attempts are made to move or manipulate the patient's head or neck.

NOTE: Rapid CT scanning is readily available in most hospitals that treat patients with head injuries; thus the routine use of CT has been advocated as a screening tool to triage patients with minor or mild head injuries who require hospital admission or surgical intervention from those who can be safely discharged without hospital admission.¹³

Cranium

TRAUMA

- Lateral, horizontal beam
- AP
- AP axial

Clinical Indications

- Calvarial fractures, penetrating injuries, and radiopaque foreign bodies

Technical Factors

- Minimum SID—40 inches (100 cm)
- IR size—10 × 12 inches (24 × 30 cm), landscape (aligned to the anterior to posterior dimension of the skull).
- Grid
- kVp range—70–85

Patient Position

- Patient supine; remove all metal, plastic, or other removable objects from head. Do not remove cervical collar unless approved by attending physician

Part Position



AP Lateral, Horizontal Beam

- If patient's head can be manipulated (see previous WARNING), carefully elevate skull on a radiolucent sponge (Fig. 15.75). If you cannot manipulate head, move patient to edge of table and then place grid IR at least 1 inch (2.5 cm) below tabletop and occipital bone, as shown in Fig. 15.76. The divergent beam then will not project the posterior skull off the IR. Place the side of interest closest to the IR.
- Place head in true lateral position by ensuring that the MSP is parallel to the IR, the image plate (IP) is perpendicular to the IR, and the infraorbitomeatal line (IOML) is perpendicular to the tabletop (see previous WARNING).
- Adjust IR to ensure that the entire skull will be included on image and center of grid is centered to CR.
- Make sure not to cut off the top of the skull. If required, use a larger IR to include the entire skull.

CR

AP Lateral, Horizontal Beam

- A **horizontal beam** (essential for visualization of intracranial air-fluid levels) is directed perpendicular to IR.
- Center to a point **2 inches (5 cm) superior to the EAM**.

Radiation Safety

- Exposure factor selection should be optimized in accordance with the ALARA.
- Collimate on four sides to anatomy of interest.
- Shield radiosensitive tissues outside area of interest.

Respiration

- Suspend respiration during exposure.

REMINDER: On a patient with a cervical spine injury, do not attempt to raise and place support under the head or to move any portion of the head or neck, as shown in Fig. 15.75, until cervical pathology has been ruled out with a horizontal beam lateral cervical.

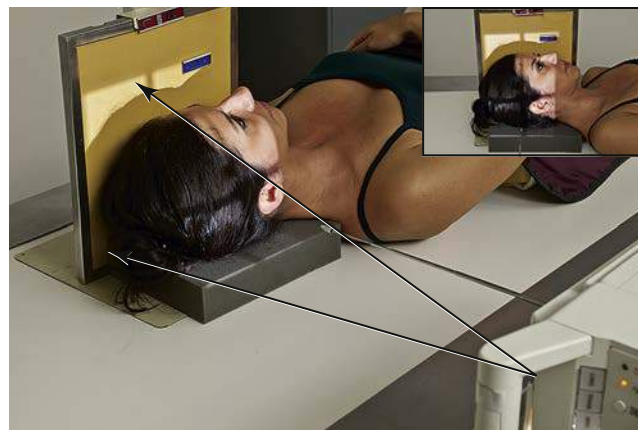


Fig. 15.75 Trauma lateral after cervical injury has been ruled out. Support is placed under elevated head.

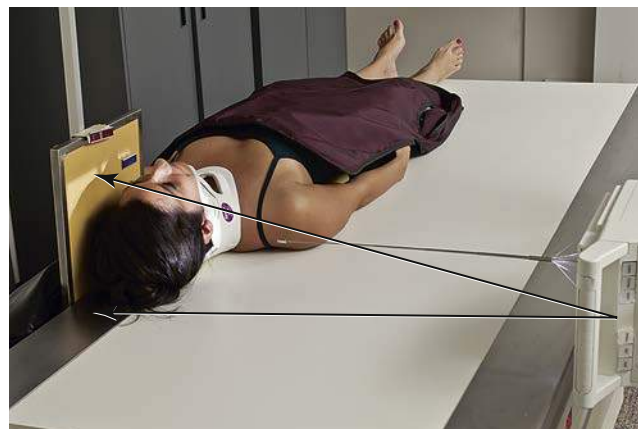


Fig. 15.76 Trauma lateral without head manipulation.

CRANIUM—AP, AP AXIAL 15° (REVERSE CALDWELL METHOD)

WARNING: Cervical spine fractures and subluxations or dislocations must be ruled out before attempts are made to move or manipulate the patient's head or neck.

NOTE: Rapid CT scanning is readily available in most hospitals that treat patients with head injuries; thus the routine use of CT has been advocated as a screening tool to triage patients with minor or mild head injuries who require hospital admission or surgical intervention from those who can be safely discharged without hospital admission.¹³

Cranium
TRAUMA

- Lateral, horizontal beam
- AP
- AP axial

Clinical Indications

- Calvarial fractures, penetrating injuries, and radiopaque foreign bodies

Technical Factors

- Minimum SID—40 inches (100 cm)
- IR size—10 × 12 inches (24 × 30 cm), portrait
- Grid
- kVp range—75–90

Patient Position

- Patient supine; remove all metal, plastic, and other removable objects from head. Do not remove cervical collar unless approved by attending physician.

Part Position



AP and AP Axial 15°

- Align MSP perpendicular to midline of grid or table (see previous WARNING)
- Center IR to CR

CR

AP to Orbitomeatal Line Projection (Fig. 15.77)

- Angle CR parallel to orbitomeatal line (OML): With patient in a cervical collar, this often occurs approximately 10° to 15° caudad, but each patient and situation will be different.
- Center CR to glabella; then center IR to projected CR.

AP Axial 15° Reverse Caldwell Method Projection (Fig. 15.78)

- Angle CR 15° cephalad to OML: To accomplish this, first find the OML on the patient; this varies in patients in cervical collars with the neck extended. Then, angle the CR 15° cephalic to the patient's OML.
- Center CR to nasion; then center IR to projected CR.

Radiation Safety

- Exposure factor selection should be optimized in accordance with the ALARA.
- Collimate on four sides to anatomy of interest.
- Shield radiosensitive tissues outside area of interest.

Respiration Suspend respiration during exposure.

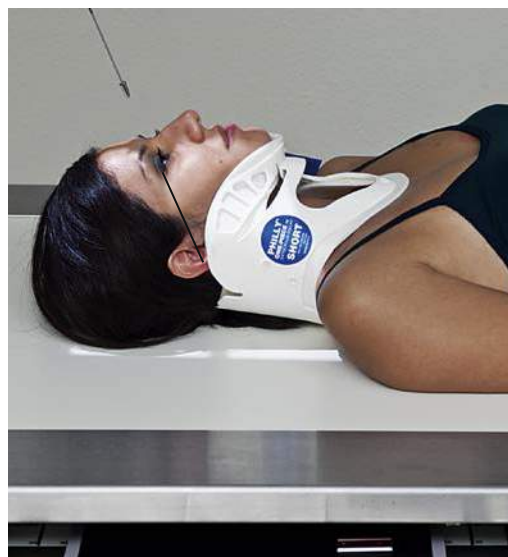


Fig. 15.77 AP CR parallel to OML, centered to glabella.



Fig. 15.78 AP axial 15° reverse Caldwell method—CR 15° cephalad to OML, centered to nasion.

CRANIUM—AP 30° AXIAL (TOWNE METHOD)

WARNING: Cervical spine fractures and subluxations or dislocations must be ruled out before attempts are made to move or manipulate the patient's head or neck.

NOTE: Rapid CT scanning is readily available in most hospitals that treat patients with head injuries; thus the routine use of CT has been advocated as a screening tool to triage patients with minor or mild head injuries who require hospital admission or surgical intervention from those who can be safely discharged without hospital admission.¹³

Cranium
TRAUMA
• Lateral, horizontal beam • AP • AP axial

Clinical Indications

- Calvarial fractures, penetrating injuries, and radiopaque foreign bodies

Technical Factors

- Minimum SID—40 inches (100 cm)
- IR size—10 × 12 inches (24 × 30 cm), portrait
- Grid
- kVp range—75–90

Patient Position

- Patient supine; remove all metal, plastic, and other removable objects from head. Do not remove cervical collar unless approved by attending physician.

Part Position



AP Axial (Towne Method) Projection

- Align MSP perpendicular to midline of grid or table (see previous WARNING)
- Center IR to CR

CR

AP Axial (Towne Method) Projection (Fig. 15.79)

- Angle CR 30° caudad to OML, or 37° caudad to the IOML. (Once again, note that a patient in a cervical collar with the neck extended will have OMLs and IOMLs that vary from the conventional parallel and perpendicular relationships formed through routine positioning; see NOTE).
- Center CR to pass **midway between EAMs** and exiting the foramen magnum. This centers CR to midsagittal plane 2¼ inches (6 cm) above superciliary arch; then center IR to projected CR.

NOTE: The CR for the AP axial should not exceed 45°, or excessive distortion will hinder the visualization of essential anatomy. If the CR cannot be angled 30° to the OML (before the maximum angle of 45° is reached), the dorsum sella and posterior clinoid processes will be visualized **superior** to the foramen magnum.

Radiation Safety

- Exposure factor selection should be optimized in accordance with the ALARA.
- Collimate on four sides to anatomy of interest.
- Shield radiosensitive tissues outside area of interest.

Respiration Suspend respiration during exposure.



Fig. 15.79 AP axial Towne—CR 30° caudad to OML, centered to midpoint between EAMs.

FACIAL BONES—LATERAL, HORIZONTAL BEAM

WARNING: Cervical spine fractures and subluxations or dislocations must be ruled out before attempts are made to move or manipulate the patient's head or neck.

NOTE: Rapid CT scanning is readily available in most hospitals that treat patients with head injuries; thus the routine use of CT has been advocated as a screening tool to triage patients with minor or mild head injuries who require hospital admission or surgical intervention from those who can be safely discharged without hospital admission.¹³

Clinical Indications

- Fractures, penetrating injuries, and radiopaque foreign bodies

Facial Bones

TRAUMA

- Lateral, horizontal beam
- Acanthioparietal (reverse Waters method)
- AP (see previous pages)

OPTIONAL

- Modified acanthioparietal (modified reverse Waters method)



Fig. 15.80 Trauma horizontal beam lateral—facial bone projection.

Technical Factors

- Minimum SID—40 inches (100 cm)
- IR size—10 × 12 inches (24 × 30 cm), portrait
- Grid
- kVp range—75–85

Patient Position

AP Lateral, Horizontal Beam

- Patient supine; remove all metal, plastic, or other removable objects from head. Do not remove cervical collar unless approved by attending physician.

Part Position



AP Lateral, Horizontal Beam (Fig. 15.80)

- Place head in true lateral position by ensuring that the MSP is parallel to the IR, the image plate (IP) is perpendicular to the IR, and the infraorbitomeatal line (IOML) is perpendicular to the tabletop (see previous WARNING).

CR

- A **horizontal beam** (essential for visualization of intracranial air-fluid levels) is directed perpendicular to IR.
- Center CR to the **zygoma**, midway between the outer canthus and EAM.

Radiation Safety

- Exposure factor selection should be optimized in accordance with the ALARA.
- Collimate on four sides to anatomy of interest.
- Shield radiosensitive tissues outside area of interest.

Respiration Suspend respiration during exposure.

FACIAL BONES—ACANTHIOPARIETAL (REVERSE WATERS METHOD) AND MODIFIED ACANTHIOPARIETAL (MODIFIED REVERSE WATERS)

WARNING: Cervical spine fractures and subluxations or dislocations must be ruled out before attempts are made to move or manipulate the patient's head or neck.

NOTE: Rapid CT scanning is readily available in most hospitals that treat patients with head injuries; thus the routine use of CT has been advocated as a screening tool to triage patients with minor or mild head injuries who require hospital admission or surgical intervention from those who can be safely discharged without hospital admission.¹³

Clinical Indications

- Fractures, penetrating injuries, and radiopaque foreign bodies

Technical Factors

- Minimum SID—40 inches (100 cm)
- IR size—10 × 12 inches (24 × 30 cm), portrait
- Grid
- kVp range—70–85

Patient Position

Acanthioparietal (Reverse Waters Method)

- Patient supine; remove all metal, plastic, or other removable objects from head. Do not remove cervical collar unless approved by attending physician.

Part Position

Acanthioparietal (Reverse Waters Method) (Fig. 15.81)

- Position MSP perpendicular to midline of grid or table (see previous WARNING).

CR

Acanthioparietal (Reverse Waters Method)

- Angle CR cephalad as needed to align CR parallel to MML.

NOTE: This projection best visualizes facial bone structures and the maxillary region by projecting the maxilla and maxillary sinuses above the petrous ridges (see arrows in Fig. 15.82).

- Center to **acanthion**; then center IR to projected CR.

Part Position

Modified Acanthioparietal (Modified Reverse Waters Method) (Fig. 15.83)

- Position MSP perpendicular to midline of grid or table (see previous WARNING).

CR

- Angle CR cephalad as needed to align CR parallel to LML.

NOTE: This projection best demonstrates the floor of the orbits and provides a view of the entire orbital rims. Petrous ridges are visualized in midmaxillary sinus region (Fig. 15.84).

- Center to **acanthion**; then center IR to projected CR.

Radiation Safety

- Exposure factor selection should be optimized in accordance with the ALARA.
- Collimate on four sides to anatomy of interest.
- Shield radiosensitive tissues outside area of interest.

Respiration Suspend respiration during exposure.

Facial Bones

TRAUMA

- Lateral, horizontal beam
- Acanthioparietal (reverse Waters method)
- AP (see previous pages)

OPTIONAL

- Modified acanthioparietal (modified reverse Waters method)



Fig. 15.81 Acanthioparietal (reverse Waters method)—CR parallel to MML, centered to acanthion.

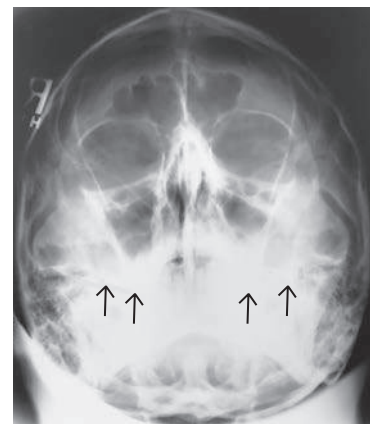


Fig. 15.82 Acanthioparietal (reverse Waters method).



Fig. 15.83 Modified acanthioparietal (modified reverse Waters method)—CR parallel to LML, centered to acanthion.

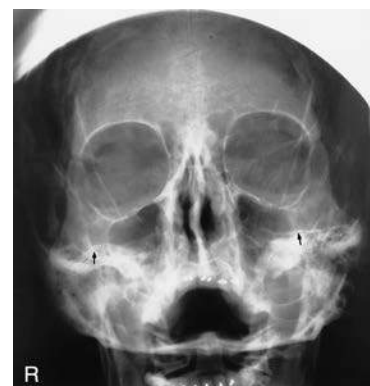


Fig. 15.84 Modified acanthioparietal (modified reverse Waters method).

SURGICAL RADIOGRAPHY

Radiography in surgery is one of the most demanding challenges encountered by a radiologic technologist. The technologist will be called on to perform procedures quickly and accurately in a sterile environment, with a minimum number of repeat exposures (Fig. 15.85). For most surgical procedures, the patient is under general anesthesia and time is of the essence, because the less time a patient spends under general anesthesia, the less likely it is that complications will occur. Therefore, the surgeon expects the technologist to perform any requested procedure without error or delay. These added pressures may create uncertainty and anxiety for the radiography student or recent graduate. However, with a solid knowledge of the surgical procedure and operation of the imaging equipment, the technologist can function effectively in the surgical suite.

Through supervised observations with an experienced surgical technologist, the student can become comfortable and confident in the surgical environment. It is essential that the student technologist be kept under the direct supervision of an experienced technologist in the OR until he or she has achieved competency for a specific procedure.

This section of the chapter identifies essential skills and commonly used equipment, and previews the surgical environment to provide the student radiographer with a baseline knowledge. More commonly performed procedures are discussed at the end of this chapter; however, it is important for student radiographers to understand that they will participate in a variety of surgical procedures based on their clinical setting. Rather than focus on specific surgical procedures, student radiographers should focus their efforts on developing the essential skills and transfer their newly acquired skills from one procedure to the next.

Essential Attributes of the Radiologic Technologist in Surgical Radiography

Although confidence and knowledge of procedures are needed in all aspects of radiography, certain personal attributes, skills, and insight are the trademark of a competent radiologic technologist in the surgical setting.

CONFIDENCE

Although no one can teach a technologist confidence, it is the first attribute that the other members of the surgical team expect to see in the technologist. **Confidence is judged by the technologist's level of comfort and ease in the OR suite, including the skilled use of imaging equipment, ability to problem-solve situations, and respect for the sterile field.** The surgical team expects the technologist to be confident in his or her abilities to perform the procedure quickly and accurately, with a minimum of repeat exposures. However, confidence comes only with experience and knowledge of all aspects of radiography. As the technologist gains experience and success in the OR, confidence will grow.

COMMUNICATION

It is essential that the technologist be an excellent communicator. He or she must communicate with other members of the surgical team regarding any concerns that arise during the procedure. **Clear communication between the technologist, surgeon, and anesthesiologist is paramount for most radiographic procedures.** For example, during operative cholangiography, the technologist must coordinate the exposure with the surgeon who is injecting the contrast medium and with the anesthesiologist who is suspending respiration. Without this team approach, motion may result and the quality of the exposure may be compromised.

The technologist must communicate radiation safety concerns with the surgical team, including failure to wear aprons, overuse of C-arm real-time imaging, and placement of hands into the radiation field. In these situations during surgery, the technologist is the radiation safety expert and must minimize exposure for the surgical team.

PROBLEM-SOLVING SKILLS

Even when the technologist has used the best knowledge and preparation, unexpected problems can occur during surgery. C-arms can cease to work, reliable exposure factors may fail to produce a diagnostic image, or the sterile field may be violated. Although it is difficult to predict every situation that might occur in the OR, the radiologic technologist **must be able to find solutions** to these problems quickly. Perhaps the most important skill of the technologist is the ability to problem-solve unforeseen situations immediately.

MASTERY

Mastery of all aspects of radiography, including use of the C-arm and mobile radiographic equipment, is essential. The technologist must be able to operate and troubleshoot conventional and digital equipment. The technologist must also know reliable exposure factors for patients of different sizes and for various procedures.



Fig. 15.85 Radiography in the surgical suite. (Courtesy GE Healthcare.)

Surgical Team

The composition of the surgical team will vary, depending on the surgeon, institutional policy, type of procedure, and other factors. A typical surgical team consists of the following members (Fig. 15.86).

SURGEON

The surgeon is a physician licensed and trained in general surgery or a specialty such as cardiovascular or orthopedic procedures. She or he has the primary responsibility for the entire surgical procedure and for the well-being of the patient prior to, during, and immediately following surgery.

ANESTHESIOLOGIST

A physician anesthesiologist or a certified nurse anesthetist specializes in administering anesthetic drugs to induce and maintain anesthesia in the patient during surgery. This person has the responsibility of ensuring the safety of the patient and monitoring physiologic functions and fluid levels of the patient during surgery.

SURGICAL ASSISTANT

A physician, physician assistant, certified surgical technologist (CST), or registered nurse (RN) assists the surgeon. This person's range of responsibilities may include suctioning, tying and clamping blood vessels, and assisting in cutting and suturing tissue.

CERTIFIED SURGICAL TECHNOLOGIST

A CST is a health professional who prepares the OR by supplying it with the appropriate supplies and instruments. Other CST responsibilities include preparing the patient for surgery and helping connect surgical equipment and monitoring devices. During surgery, CSTs have the primary responsibility for maintaining the sterile field.

CIRCULATOR

A circulator is a nonsterile CST or RN who assists in the OR by responding to the needs of scrubbed members in the sterile field before, during, and after the surgical procedure. Duties may include recording of pertinent information, retrieval of additionally needed items, and connecting nonsterile surgical equipment.

SCRUB

A scrub is a CST or RN who prepares the sterile field scrubs, gowns the members of the surgical team, and prepares and sterilizes the instruments before the surgical procedure is begun (Fig. 15.87).

NOTE: During OR cases, the radiologic technologist receives instructions from a physician (surgeon, anesthesiologist).



Fig. 15.86 Surgical team—surgeon, certified surgical technologist (CST), and radiologic technologist discussing procedure with patient.



Fig. 15.87 Scrub preparing and maintaining sterile surgical field.

Surgical Radiography Imaging Equipment

The technologist must be familiar with the location of power outlets to be used for a procedure. Ideally, all imaging equipment should be in place and checked for correct operation before the procedure is begun. Although most surgical equipment remains in the surgical area, it must be cleaned and checked frequently for correct operation. Once the procedure has begun, there is no time to troubleshoot equipment or fix problems.

Daily, weekly, and monthly quality control protocols should be followed for all surgical radiographic equipment. Even a small problem, such as a frayed electrical cord, must be addressed before it results in an equipment failure.

CLEANING

Mobile (portable) units and C-arm equipment should be cleaned before and after use in the surgical area. An approved antiseptic cleaner should be used to wipe down the equipment. A liquid-type cleaner rather than an aerosol is recommended to prevent the introduction of airborne contaminants into the surgical area. The technologist must wear gloves when cleaning equipment, especially if blood or other body fluids are present.

Equipment permanently stored in the surgical area must be cleaned weekly or as needed. IRs and grids must be inspected for contamination and cleaned weekly.

OPERATIONAL CHECK

Before imaging equipment is used, an operational check should be performed. A log of any problems and failures should be maintained and monitored.

PROPER EQUIPMENT LOCATION

The technologist must be familiar with the location of power outlets (and data ports for uploading images to a picture archiving and communications system [PACS]) to be used for a procedure. Ideally, all imaging equipment should be in place and checked for correct operation before the procedure.

If C-arm fluoroscopy units are being used, place monitors in clear vision of the surgeon. Make sure that placement of the C-arm or portable unit is not interfering with normal foot traffic.

MOBILE C-ARM DIGITAL FLUOROSCOPY SYSTEMS

Another type of mobile imaging equipment is the C-arm mobile fluoroscopy system. The term *C-arm* describes a basic design of a mobile fluoroscopy unit, which forms a large C shape with the x-ray tube located at one end of the C-arm and the image intensifier tower at the other (Fig. 15.88).

Familiarity with the C-arm, monitor, and image controls is essential for the technologist who is performing ED or procedures, during which these systems are most commonly used. One also must become familiar with the various types of special beds or carts used with the C-arm. For example, a surgical bed may not accommodate the C-arm x-ray tube under the table in the abdominal area because of the base supports unless the patient's head is placed at the correct end of the bed or cart (Fig. 15.89).



Fig. 15.88 OEC 9900 Elite. (Courtesy GE Healthcare.)



Fig. 15.89 OEC 9900 Elite. (Courtesy GE Healthcare.)

Maneuverability

The equipment is designed to be very maneuverable. The C-arm itself is attached to a beam located on the base of the C-arm that can be raised, lowered, or extended as needed. The base of the C-arm and the supporting beam provide a much-needed counterbalance to the C-arm portion. This counterbalance stabilizes the machine during any needed rotation, tilt (cephalic or caudal angles), or horizontal-beam, projection (Fig. 15.90). The C-arm can also be rotated 180° to place the tube on top and the intensifier on the bottom. However, this is not recommended because of the increase in OID, which decreases image resolution and increases scatter radiation. The tube-on-top position also results in a significant increase in exposure to the eyes, head, and neck areas of the surgeon or radiologist because of the exposure pattern of the C-arm in this orientation (see Fig. 15.97).

Overall, the unit is flexible to use. The technologist must be familiar with a variety of built-in joints, extensions, and adjustments. With its three-wheel base, steerable rear wheels, and a swiveling nose wheel, the operator can easily maneuver the unit into almost any possible configuration, with reasonable space.

Display Monitors and Control Cart

Two monitors are generally used, so the active image can be displayed on one monitor while the second monitor can be used to hold an image for reference purposes. Generally, the active monitor is on the technologist's left and the hold monitor is on the right. Images also can be rotated or flipped as needed for preferred viewing by the surgeon and/or radiologist.

Uses of C-Arm

The technologist will use the C-arm unit with various types of procedures in which mobile fluoroscopy and/or still frame imaging is needed. Common surgical procedures may include cholangiography, open or closed reductions of fractures, and hip pinning.

Images can be stored temporarily by video memory or on hard disks. With the advancement and popularity of the PACS, images can be directly uploaded to a PACS with the appropriate data connection. Optional hard copy printers are also available for printouts. Cine loop capability, wherein images are recorded in rapid succession while contrast medium is injected and then displayed as a moving (or cine) image, is possible.

As with other types of digital imaging, image enhancement and manipulation are possible, including overall brightness and contrast controls, magnification, edge enhancement, masking, and digital subtraction studies. These manipulations can be made during fluoroscopy or for postimage processing, depending on the manufacturer.

Controls and Operation Modes

The digital C-arm fluoroscopy systems include a variety of operating mode control options with which the technologist must be familiar. These control panels may be located on the display monitor control cart, on the C-arm unit itself (Fig. 15.91), or on an attached or detached remote control.

The magnification mode is the ability of a system to magnify the image for better visualization of structures because surgeons frequently need to view the image at a specified distance from the monitor.

The pulse mode is used to create an x-ray beam that pulsates at timed increments to reduce exposure.

The snapshot or digital spot mode activates a digital spot, which results in a higher quality computer-enhanced image as compared with a held fluoroscopic image.

Auto/manual exposure control allows for exposure by the operator if desired, or the use of automatic exposure control (AEC).

Additional optional modes available on some equipment that allow more complicated procedures include subtraction (digital subtraction) and roadmapping. Subtraction is a technique in which an initial image is recorded during continuous fluoroscopy. The initial image then is used as a filter for the next fluoroscopic images. Essentially, the C-arm subtracts the initial image from all the other images produced. All stationary structures are removed (subtracted) from the image, and only moving (or new/different) structures are imaged. When fluoroscopy ends, the C-arm resets back to normal mode. For example, subtraction is sometimes used for operative cholangiography. Use of subtraction will eliminate the stationary ribs, spine, and surrounding soft tissues, leaving only the moving injected contrast medium to be imaged. Therefore, the final images show the contrast-filled biliary system free from superimposition of the surrounding structures.

Roadmapping is a method of image display wherein a specific fluoroscopic image is held on the screen in combination with continuous fluoroscopy. It is similar to subtraction in that it removes stationary structures from the viewing screen. This is especially useful in interventional procedures that require the placement of catheters.



Fig. 15.90 Horizontal setup for lateral hip.



Fig. 15.91 Control panels of OEC 9600 ELITE series C-arm. (Courtesy GE OEC Medical Systems, Salt Lake City, UT.)

Foot Pedal

The foot pedal allows the physician or other operator to have hands-free operation of the C-arm. A fully equipped foot pedal has multiple controls for various functions, as shown in Fig. 15.92.

Image Orientation

The flexibility of the C-arm for imaging a variety of anatomic structures at almost any conceivable angle from any side or direction requires correct image orientation each time it is set up for use. This should be completed during setup time before the initial image is obtained to avoid needless exposure to the patient and personnel.

Because of the variety of policies and procedures reported in various hospitals and medical centers, technologists must develop their own methods of achieving correct image orientation before the patient is imaged. One method is to bring the C-arm into the room in the same position and orientation that will be used for the procedure. Place an R lead marker on the flat surface of the x-ray tube collimator, oriented in the same way the patient will be oriented. The top of the letter R should be to the head end, indicating the patient's right side, to appear anatomically correct on the monitor to the viewer's left. (This is the same orientation as for viewing radiographs; that is, the patient's right to the viewer's left.) At this point, the image can be flipped or rotated as needed to produce the correct orientation necessary for the procedure. An exposure can be provided during this setup with an apron or other shielding covering the C-arm to shield other personnel in the room. Correctly viewing and orienting the test image of the R on the monitor is an important preparation for the procedure.

O-ARM

The O-arm is a new type of surgical imaging equipment that a radiologic technologist may operate during select procedures (Fig. 15.93). The O-arm imaging system is a mobile x-ray system designed for 2D fluoroscopic and 3D imaging and is intended to be used where a physician benefits from 2D and 3D information of anatomic structures and objects with high x-ray attenuation such as bony anatomy and metallic objects. The O-arm imaging system is compatible with certain image-guided surgery systems.¹⁴

The O-arm is an example of new technology that can be operated in the surgical setting by a radiologic technologist; however, the C-arm continues to be the primary imaging equipment with fluoroscopic capabilities used, and therefore, the C-arm is the focus of this section.

Radiation Protection in Surgical Radiography

Conscientious radiation protection practices are especially important in mobile radiography and surgery suites in which fixed protective barriers do not provide a shielded place to stand during exposures. This is true with all mobile x-ray examinations, but even more so with C-arm mobile fluoroscopy, which potentially results in considerably more scatter radiation to the immediate area over a longer time. The technologist must continually be aware of the three important cardinal rules of radiation protection—time, distance, and shielding.

STAFF RADIATION EXPOSURE

The primary source of radiation exposure to the fluoroscopy staff is from scattered radiation from the patient. Scattered x-rays originate from the volume of tissue irradiated by the x-ray beam (structures within the field of view). Nearby radiation levels depend on C-arm orientation, technique factors, and patient size, but decrease rapidly with distance from the patient.

The operator and all who remain in the room during the exposure should always wear a lead apron. A 0.5-mm lead equivalent

apron, which reduces the exposure by a factor of 50% or more over the diagnostic x-ray energy range, is recommended.¹⁵ A wrap-around apron is necessary if the person's back is repeatedly turned toward the patient. In addition, staff members should always move away from the patient during x-ray beam activation if their immediate presence is not required. Extra care must be taken to ensure that all personnel are conscientious about the location of the C-arm and each adjusts their position appropriately so that the lead shield remains between themselves and the C-arm. In addition, the operator, even if wearing a lead apron, should always stand a minimum of 6 feet (2 m) from the x-ray tube during all exposures.



Fig. 15.92 Foot pedal controls.



Fig. 15.93 O-arm imaging system. (Reprinted with the permission of Medtronic, Inc. © 2017.)

C-ARM ORIENTATION AND EXPOSURE PATTERNS

Vertical Posteroanterior Projection CR

If it is assumed that the patient is supine, keeping the C-arm posteroanterior (PA; the fluoroscopy tube is located below the patient, and the image intensifier is located above the patient) and directly perpendicular to the floor minimizes exposure to the neck and facial region (Fig. 15.94). If the C-arm is tilted as much as 30°, as shown in Fig. 15.95, the configuration of the exposure fields changes to increase exposure significantly to the upper body and facial region not shielded by the lead apron. **Studies have shown that even a 30° C-arm tilt will increase the dose to the face and neck region of an operator of average height who stands next to the C-arm by a factor of 4.**

Horizontal Projection CR

The configuration of the exposure fields with a horizontal beam is demonstrated in Fig. 15.96. Note that the **exposure region on the x-ray tube side of the patient is significantly larger** than that in the region near the intensifier tower. This should be an important consideration for the surgeon or other operator who may need to stay near the patient.

Vertical AP Projection CR

Occasionally, the technologist may be asked to reverse the C-arm, tube on top and image intensifier on bottom. This may provide the surgeon more room for manipulation; however, **this is not recommended due to a significant increase in exposure to the operator**, as shown in Fig. 15.97 (up to 100 times higher dose to the eyes of the operator).

Patient Dose

Modern fluoroscopic units produce images with an image intensifier (or flat panel detector) that captures the x-rays exiting the patient. The detector converts the time sequence of incident x-rays to a series of images displayed on the monitor. With manual technique, the image quality and brightness are adversely affected when the operator pans across tissues of different thickness and composition. For this reason, fluoroscopy is almost exclusively performed using automatic brightness control (ABC). Technical factors (e.g., kVp, mA, filtration, and/or pulse length) are then adjusted automatically to maintain image brightness at a constant, proper level.

The exposure rate depends on operational mode, field size, patient size, tissue composition, and ABC design. Attenuation of x-rays along the beam path influences the number of x-rays reaching the detector. The ABC compensates brightness loss caused by lower transmission of x-rays through the patient by generating more x-rays and/or producing more penetrating x-rays (reducing image contrast). The maximum exposure rate at 1 foot (30 cm) from the image receptor (IR) cannot exceed an air kerma rate of 88 mGy/min, which equals an exposure rate of 10 R/min. In high-level fluoroscopy (HLF) mode, the maximum exposure rate at the same reference point cannot exceed 20 R/min, corresponding to an air kerma rate of 176 mGy/min. Generally, the entrance exposure rate to the patient is 1 to 3 R/min, but the ABC can boost the exposure rate to maximum for large patients.

Digital spot film acquires a single, static radiographic image of the structures of interest with a low radiation dose. This is an excellent dose reduction method to evaluate static spatial relationships or document the proper location of a device.

Gonadal shielding should be applied when the x-ray beam is directed toward the abdomen and pelvis if the presence of such shielding does not interfere with the examination.

Summary of Radiation Protection in Surgical Radiography

Sound radiation protection practices are important for protection of all personnel during mobile imaging, as already described. A summary of what this includes in the surgical setting follows.

TIME

Use of Intermittent Fluoroscopy

- Single-exposure capacity can greatly reduce fluoroscopy time.
- The image hold feature allows the last image to remain on the monitor.

Minimize Boost Exposures

The boost feature on most C-arms provides an improved and brighter image for the patient with a large body habitus, or thick anatomy. However, this feature increases radiation, primarily mA, which also increases exposure to the patient and surrounding surgical team by a factor of three to four times compared with standard fluoroscopy. Use the boost feature only when no other alternative or adjustment will improve the image.

DISTANCE

Vertical Alignment

Place the vertical alignment of the C-arm so the x-ray tube is away from the operator's head and neck region. This is achieved by placing the x-ray tube beneath the OR table, thus reducing the dose to the head and neck region of the surgical team. (See p. 604 for details of C-arm orientation and exposure patterns.)

Minimize Distance between Anatomy and Image Receptor

Reducing the distance between the anatomy and the image intensifier creates a brighter, sharper, and less magnified image, with a reduction in radiation to the immediate area.

SURGICAL TEAM RADIATION PROTECTION SHIELDING

Protective Lead Aprons

- Provide aprons for all personnel.
- Wear a thyroid collar.
- Secure aprons tightly to prevent them from touching the sterile field or sterile personnel.
- Clean aprons weekly or as needed with a liquid-type cleaner.
- Intermittently check aprons for cracks in the lead lining.

COMMUNICATION

Coordination of Exposure with Surgical Team

Coordinate exposures among the anesthesiologist, surgeon, and surgical team. For studies such as operative cholangiography, injection of contrast media, and suspension of patient breathing, x-ray exposure must be closely coordinated among team members.

The technologist should clearly announce "x-ray" or "x-ray on" before initiating an exposure to enable nonessential staff to leave the area or get behind lead shields. Announcing "x-ray off" communicates to nonessential staff that it is safe to return to the area.

Monitor Personal Dosimetry Report

Technologists who frequently perform C-arm procedures should closely monitor their personal dosimetry. Personnel dosimeters must be worn at collar level outside of protective apron. If they discover excessively high levels, they may have to modify their work habits and discuss strategies to reduce dose levels with the department radiation safety officer.

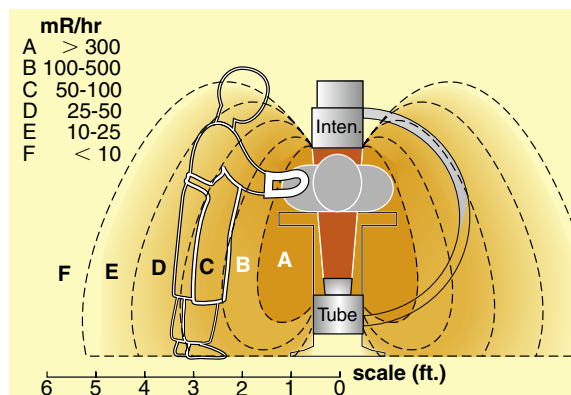


Fig. 15.94 Exposure levels—CR vertical, PA projection, intensifier on top (least exposure to operator).

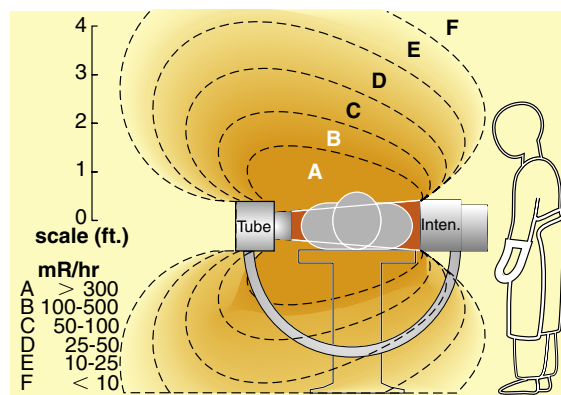


Fig. 15.96 Exposure patterns and levels—CR horizontal (least exposure at intensifier side). (Adapted from Technical reference, Salt Lake City, Utah, 1996, OEC Medical Systems; and Geise RA, Hunter DW: Personnel exposure during fluoroscopy procedures, *Postgrad Radiol* 8:162–173, 1988.)

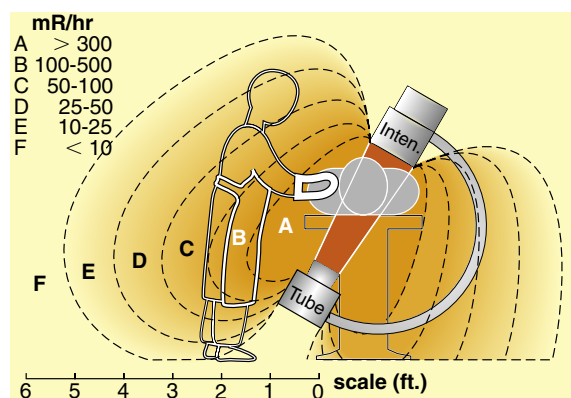


Fig. 15.95 Exposure levels—CR 30° from vertical (CR angle increases exposure to operator). (Adapted from Technical reference, Salt Lake City, Utah, 1996, OEC Medical Systems; and Geise RA, Hunter DW: Personnel exposure during fluoroscopy procedures, *Postgrad Radiol* 8:162–173, 1988.)

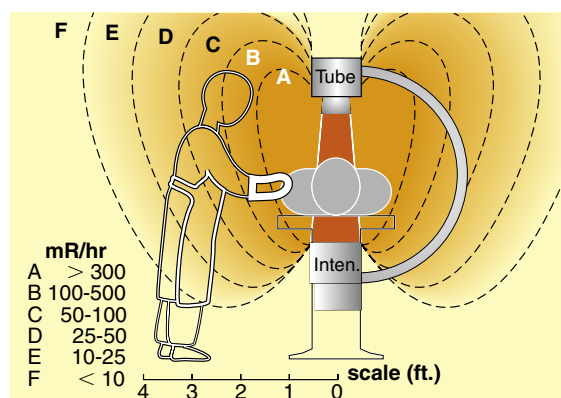


Fig. 15.97 Exposure levels, AP projection (tube on top) (not recommended). (Adapted from Technical reference, Salt Lake City, Utah, 1996, OEC Medical Systems; and Geise RA, Hunter DW: Personnel exposure during fluoroscopy procedures, *Postgrad Radiol* 8:162–173, 1988.)

Surgical Asepsis Principles

Unfortunately, it is impossible to remove all infectious organisms from the OR. Surgical asepsis consists of the practice and procedures used to **minimize the levels of infectious agents present in the surgical environment**. Through the use of safe practices, wearing proper surgical attire, and ensuring that care is taken around the surgical incision, the patient's exposure to these infectious agents is greatly minimized. This requires a clear separation of sterile items and areas from nonsterile areas in the surgical suite (Fig. 15.98).

To reduce the risk for infection of the patient during surgery, the following principles of surgical asepsis must be followed:

1. Only sterile items are allowed within the sterile field.
2. If the sterility of an object is in doubt, it must be considered nonsterile.
3. If a sterile drape or cover is touched by a nonsterile object or person, it must be considered contaminated.
4. Nonsterile personnel must not come into contact with a sterile barrier, drape, surgical instrument, or sterile personnel.
5. Any contaminated sterile drape or cover must be reported and replaced by sterile personnel.
6. Sterile gowns are considered sterile from the shoulder to the level of the sterile field, and at the sleeve from the cuff to just above the elbow.
7. OR tables are considered sterile only at the level of the tabletop.
8. Only sterile personnel can touch sterile items.

Surgical Suite Environment

The typical surgical suite has two general regions known as **sterile** and **nonsterile** areas.

The **sterile** area includes the patient, surgical field, surgeon and surgical assistants, surgical equipment, tables, and carts. (In some facilities, the sterile area includes the area surrounding the sterile field, up to 1 foot wide.) Often, most of the sterile area is located to one side of the room, leaving the other side of the room available for necessary nonsterile personnel. **The technologist and the imaging equipment must not violate the sterile area.** Student technologists new to the OR must have a clear understanding of the differences between sterile and nonsterile areas. When in doubt, ask the radiologic technologist, CST, or circulator for clarification. If the sterile area is violated, which may contaminate the instruments used for the procedure, the technologist must report this event immediately. Because the violation may have not been noticed by the surgical team, the technologist has a critical responsibility to report it. In most cases, additional sterile drapes or a new set of sterile instruments can be used to create a safe and sterile environment again.

The **nonsterile** area is where the technologist and other nonsterile surgical personnel, such as the anesthesiologist and the circulator, is located. The technologist can safely stand and operate imaging equipment within this area. For select procedures, a plastic drape or shower curtain may be erected to indicate the dividing point between sterile and nonsterile areas.

IMAGING EQUIPMENT/PERSONNEL AND STERILE FIELDS

C-Arm

C-arm use in surgical settings requires special attention in maintaining sterile fields. Vertical alignment with the intensifier on top often causes it to be placed over open incisions. Three basic approaches are commonly used to maintain a sterile field.

The first (and most commonly used) method involves draping the image intensifier, x-ray tube, and C-arm with a sterile cloth and/or bags, with a tension band or adhesive tape holding the cloth or

plastic cover in place (Fig. 15.99). Another type of image intensifier cover, called a snap cover, has a band that makes a snapping sound when it is released into position (Fig. 15.100). These types of covers also make it possible for the technologist (with guidance from the surgeon) to position the image intensifier precisely as needed over the sterile surgical site for correct centering.



Fig. 15.98 Surgical asepsis—separation of sterile and nonsterile areas.

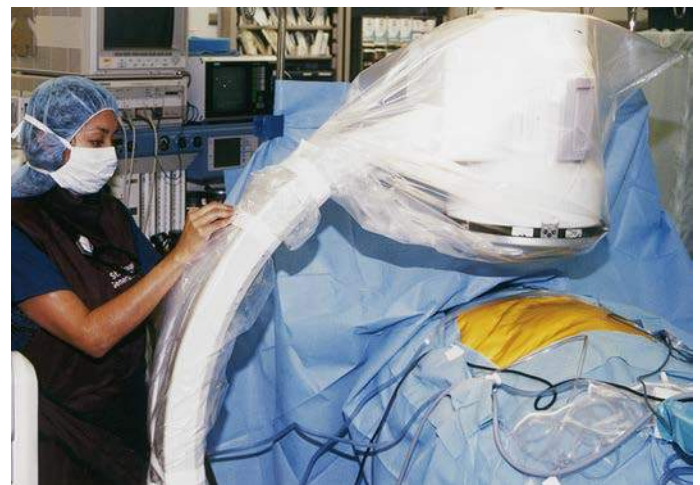


Fig. 15.99 Draping the C-arm with sterile plastic C-arm cover.



Fig. 15.100 Draping the C-arm and intensifier with snap cover. (Courtesy Philips Healthcare, Andover, MA.)

A second approach is to drape the patient (or surgery site) temporarily with an additional sterile cloth before the undraped C-arm is positioned over the anatomy. Once a satisfactory image has been obtained and the C-arm removed, the sterile cloth (or drape) is then removed from the patient and discarded. This process is repeated with a new (unused) sterile cloth if it becomes necessary to use the C-arm again. This approach is used in cases in which the physician does not need to interact with the surgical site during fluoroscopy or when snap covers are not available.

The third method of maintaining a sterile area uses a shower curtain. Hip pinnings or femoral roddings require frequent manipulation of the C-arm between PA and lateral projections to the surgical incision, making these procedures ideal for the shower curtain. A long horizontal metal bar attached to two vertical suspending rods is placed along the lateral longitudinal axis of the affected side (Fig. 15.101). A large, sterile, clear plastic sheet (called a shower curtain) is suspended from the horizontal bar, which is positioned about 3 feet (1 m) above the patient. A special opening in the middle of the plastic is attached with a second adhesive strip to the lateral aspect of the hip–proximal femur and is used for access to the incision. The curtain forms a sterile barrier between the physician and patient as the C-arm is positioned for a standard PA and horizontal beam lateral hip from the nonaffected side of the patient.

IMAGE RECEPTORS

When an IR must be used within the sterile field, it must be placed in a sterile plastic cover. Keep in mind that **only the outer surface of the cover is sterile**. The inner surface of the cover is nonsterile and comes in contact with the IR. The procedure for placing and removing an IR in a sterile cover is as follows:

1. Sterile surgical personnel hold the plastic cover open, with the top cover folded over to maintain sterility of the outer surface and their gloved hands.
2. The technologist carefully slips the IR into the cover, ensuring that the IR touches only the inner surface of the plastic cover (Fig. 15.102).
3. Surgical personnel wrap the top of the cover over and secure it.
4. The surgical staff, with verbal directions from the radiologic technologist, places the covered IR in the necessary location and the exposure is taken.
5. Once the exposure has been taken, the surgical staff removes the covered IR and hands it to the radiologic technologist.
6. The technologist removes the IR from the plastic cover by sliding the IR onto a nonsterile table or surface, with care taken not to transfer any possible body fluids from the outer cover of the plastic bag, to dispose of the IR cover in the appropriate receptacle, and to remove gloves (Fig. 15.103).
7. The image is processed.

NOTE: The technologist must wear nonsterile gloves when handling the cover because of possible blood or other body fluid exposure.



Fig. 15.101 Shower curtain, view from the technologist's (nonsterile) perspective



Fig. 15.102 Procedure for placing IR into sterile cover.



Fig. 15.103 Removing IR from sterile cover.

Surgical Attire

The technologist must change from normal work dress into the appropriate surgical attire before entering the OR. Because the technologist's typical uniform may pose a health concern for the operative patient, proper surgical attire must be worn in all restricted and nonrestricted areas in surgery (Fig. 15.104). Proper surgical attire includes the following items.

SCRUBS

Even if surgical scrubs are normally worn in the general radiology department, the radiologic technologist must change into approved surgical scrubs. Surgical scrubs should be made of a low lint-producing material, which minimizes bacterial shedding. Two-piece scrubs should fit properly, with the top tucked in at the waist. The pant legs of the scrub bottoms must not drag on the floor. In some facilities, scrub jackets are also available. Surgical scrubs must be changed following the procedure and laundered by the hospital. If soiled with blood, perspiration, or food, scrubs must be changed before the wearer reenters the surgical suite.

SCRUB COVER

Scrub covers are button-up or snapped covers worn by the technologist between procedures. They are designed to prevent soiling or cross-contamination of the scrubs while the technologist is outside the surgical suite. Scrub covers must be removed before the technologist enters the surgical suite.

HEAD COVER

A proper-fitting head cover must be worn before entry into a surgical area. The bouffant and hood types of covers are preferred because they cover the head best. All hair must be tucked inside the head covers. The hood type of cover must be worn by the technologist with a beard or other facial hair. Head covers must be discarded immediately after use and changed for each procedure.

SHOE COVERS

Shoe covers are designed to keep the shoes clean and decrease the quantity of soil and bacteria tracked into the surgical suite. They must be changed if they become soiled or torn. They should be worn even in the presurgical and recovery areas.

SHOES

Because of the volume of fluid and presence of sharps in the OR, soft cloth shoes should not be worn. A durable shoe with plenty of support and closed hard toe and heel will minimize injuries caused by falling objects, needles, and image receptors.

MASKS

A surgical mask must be worn to reduce the dispersal of microbial droplets from the technologist during surgery. Masks also will reduce the risk that pathogenic organisms present in the surgical suite may be inhaled by the technologist. A single high-filtration mask is recommended for most procedures. This mask has a pliable nose stripe and two sets of ties to secure it. The nose stripe provides a contoured fit for the wearer and helps prevent fogging for eyeglass wearers. **Masks must be changed between procedures or when moisture is detected on the outside of the mask.**

PROTECTIVE EYEWEAR

If the technologist is present during a procedure in which blood, body fluids, or tissue debris may strike the eye region, Occupational Safety and Health Administration (OSHA)-approved protective eyewear must be worn. However, this equipment will not be necessary for most of the radiographic procedures performed in surgery. In

the angiography suite, specialized lead eyewear is sometimes worn to protect the eyes of the wearer from long-term exposure to the x-ray field.

NONSTERILE GLOVES

When handling contaminated IRs or soiled IR covers, or when cleaning equipment after procedures have been performed, the technologist must wear nonsterile gloves. Once the gloves are removed, the hands must be washed.



Fig. 15.104 Surgical attire—scrubs, mask, shoe covers, head cover, nonsterile gloves, and protective apron.

SURGICAL RADIOGRAPHIC PROCEDURES

Biliary Tract Procedures**OPERATIVE (IMMEDIATE) CHOLANGIOGRAPHY****Overview of Procedure**

First performed in 1932, the operative or immediate cholangiography is performed during surgery to demonstrate anatomy of the biliary ductal system, drainage into the duodenum, and any residual stones in the biliary ducts. In many cases, the patient has a previous history of gallstones and the surgeon may be concerned about residual stones remaining undetected in one of the biliary ducts.

The operative cholangiography may be performed before or following surgical removal of the gallbladder. The surgeon places a small catheter into the biliary ducts and injects iodinated contrast media into the ducts. Once respiration has been suspended, the technologist initiates exposure and produces images using mobile radiographic or C-arm equipment.

Equipment Used and Setup

C-arm Digital Fluoroscopic Cholangiography The C-arm should be set up and correctly oriented prior to the beginning of the procedure, with the monitors set up in clear view of the surgeon. The C-arm will be positioned in vertical alignment, with the x-ray tube beneath the table. A sterile drape or cover must be placed over the image intensifier, as shown in Fig. 15.105. Make sure that the C-arm is moved away from the surgical field until needed, and then place the image intensifier over the anatomy to be imaged. Coordinate all exposures with the surgeon and anesthesiologist.

Mobile Radiographic Cholangiography A conventional mobile radiographic unit also can be used for this procedure and should be brought into the surgical suite and positioned carefully near the surgical field. Once the surgical incision has been draped by the surgeon, the x-ray tube is brought in and centered over the anatomy. Often, the surgeon will indicate the centering point with a twist or mark on the sterile towel (Fig. 15.106). The IR is placed in the IR holder (“pizza pan”) and is placed into a special slot under the surgical table that allows the IR and IR holder to slide under the surgical table until the appropriate location is reached. The IR and holder are placed into the table near the end of the table closest to the anesthesiologist. With the use of a handle, the IR is advanced until it is centered over the right upper quadrant of the abdomen. With an IR of 14 × 17 inches (35 × 43 cm), the top of the image receptor is just below the right axilla.



Fig. 15.105 C-arm-guided operative cholangiography.



Fig. 15.106 Conventional mobile x-ray unit positioned for AP projection (centering point indicated by surgeon). (Courtesy GE Healthcare.)

LAPAROSCOPIC CHOLECYSTECTOMY

Laparoscopic cholecystectomy provides a less invasive approach for the removal of diseased gallbladders. The surgeon makes a small opening in the umbilicus and passes an endoscope into the abdominal cavity. This type of procedure has long been used for visual assessment of the abdomen to detect signs of pathology or trauma. It is referred to as a laparoscopic (*lap'-ah-ro-skop'-ik*) procedure. This technique has been modified so that cholecystectomy and cholangiography can be performed with a minimal amount of surgical trauma to the patient.

Advantages of Laparoscopy

There are three advantages of laparoscopy:

1. It can be performed as an outpatient procedure.
2. It is a minimally invasive procedure. Previous surgical techniques required the creation of a large opening to remove the gallbladder. This degree of invasive surgery required the patient to remain in the hospital for at least 2 days.
3. It involves a shorter hospital stay than other procedures, with reduced cost. Many patients who undergo the laparoscopic technique can return home the same day and, in some cases, can return to work in 2 to 3 days.

The laparoscopic cholecystectomy, however, is not suited for every patient. More complex disease processes or involved procedures may require the more traditional surgical approach.

Summary of Procedures for Operative and Laparoscopic Cholangiography

1. The technologist changes into surgical attire and ensures that the portable unit or C-arm is functional and clean. The C-arm should be set to the cine loop function, if available.
2. If a portable unit is used, a scout image is taken before the patient is surgically prepped. The distance that the IR is advanced from the head of the table is noted. A special ruler and tray setup may be used in positioning the IR.
3. The scout image is processed and exposure factors adjusted with the IR positioned correctly.
4. Once the surgeon places the catheter into the biliary ducts, the portable unit is again readied for another image or the C-arm is placed over the desired anatomy. Once the radiologic technologist is prepared to obtain images, and surgeon injects contrast media.
5. Images are obtained with the cooperation and synchronization of the surgeon, anesthesiologist, and technologist. The anesthesiologist controls the breathing of the patient. Some surgeons prefer to inject all the contrast and image after injection; others prefer to view the contrast filling the biliary ducts, moving to the hepatopancreatic sphincter, and spilling into the duodenum.
6. If the OR table is tilted for the oblique positions and a mobile unit is being used, the grid and IR are placed landscape (cross-wise) to avoid objectionable grid cutoff.
7. Images are processed and may have to be reviewed by a radiologist. The technologist may convey a written report from the radiologist to the surgeon.

Images Obtained

At least two and preferably three radiographic images are obtained in slightly different positions. Each exposure is preceded by a fractional injection of contrast medium. Positions may include an **AP**, a **slight right posterior oblique (RPO)**, and a **slight LPO**. The RPO is helpful in projecting the biliary ducts away from the spine, especially with a hyposthenic patient.

The C-arm may have to be tilted to project the biliary ducts away from the spine.

Anatomy Demonstrated

Contrast-enhanced biliary ducts, including the common bile duct, hepatic ducts, and cystic ducts, are shown (Fig. 15.107). If the cholangiography is performed before the gallbladder is removed, the gallbladder also will be enhanced. If stones or biliary duct stenosis is present, the opacity of the biliary ducts will be restricted.

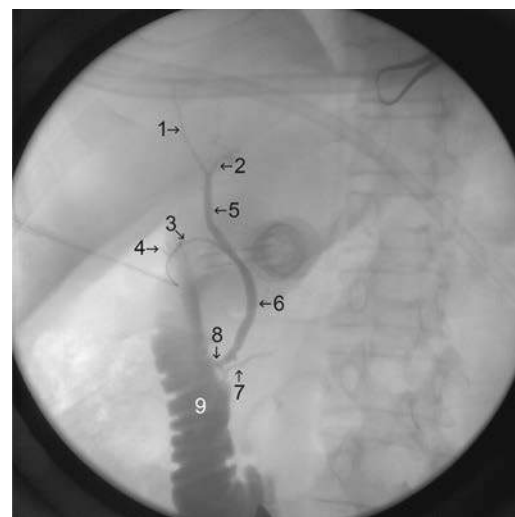


Fig. 15.107 Normal intraoperative cholangiogram. (1) Right hepatic duct; (2) left hepatic duct; (3) cystic duct; (4) two surgical clips holding the cholangio catheter in the cystic duct; (5) common hepatic duct; (6) common bile duct; (7) pancreatic duct; (8) ampulla of Vater; (9) duodenum. (From Massarweh NN, Flum DR. Role of intraoperative cholangiography in avoiding bile duct injury. *Journal of the American College of Surgeons* 204(4):656–664.)

Urinary Tract Procedures

RETROGRADE UROGRAPHY

Overview of Procedure

Retrograde urography is a nonfunctional examination of the urinary system during which contrast medium is introduced directly retrograde (backward, against the flow) into the pelvicalyceal system via catheterization by the urologist during a minor surgical procedure. Retrograde urography is nonfunctional because the patient's normal physiologic processes are not involved in the procedure. This procedure is frequently performed to determine the location of undetected calculi or other types of obstruction in the urinary system. The procedure may also be performed to study the renal pelvis and calyces for signs of infection or structural defect.

Equipment Used and Setup

The procedure usually is done as outpatient surgery in a dedicated urography room. The urography room generally contains a combination cystoscopic-radiographic table, which includes a dedicated x-ray tube (which may also house fluoroscopic capabilities), with a bucky tray built into the table. If such a table is not available, a mobile radiographic unit or C-arm may be used to image the urinary system. The patient is usually sedated or anesthetized for this examination. If the unit does not have fluoroscopy, conventional film-screen or digital imaging receptors are used for creating radiographs.

Summary of Procedure

1. The technologist changes into surgical attire and ensures that the unit (or portable unit or C-arm) is functional and clean.
2. The patient is placed in the modified lithotomy position, with the legs placed in stirrups, as illustrated in Fig. 15.108.
3. The urologist inserts a cystoscope through the urethra into the bladder. After examining the inside of the bladder, the urologist inserts ureteral catheters into one or both ureters. Ideally, the tip of each ureteral catheter is placed at the level of the renal pelvis.
4. After catheterization, a scout radiograph is exposed. The scout radiograph allows the technologist to check technique and positioning and the urologist to check catheter placement. Center to the level of the iliac crest when using a 14 × 17-inch (35 × 43-cm) IR, portrait alignment.
5. The second radiograph in a common retrograde urographic series is a **pyelogram**. The urologist injects 3 to 5 mL of contrast media directly through the catheter into the renal pelvis of one or both kidneys. Respiration is suspended immediately after injection, and the exposure is made.
6. The third and final radiograph in a common series is a **ureterogram**. The head end of the table may be elevated for this radiograph. The urologist withdraws the catheters and simultaneously injects contrast material into one or both ureters. The urologist indicates when the exposure should be made.

This examination is used to visualize the internal structures of one or both kidneys and ureters directly.

Anatomy Demonstrated

A retrograde **urogram** (Fig. 15.109) on the right side, with catheter in place, best demonstrates the renal pelvis and contrast-filled major and minor calyces. The left side shows the left ureter after the left catheter has been withdrawn; therefore, this is called a **ureterogram**.



Fig. 15.108 Modified lithotomy position for retrograde urography. (From Long BW, Rollins J, Smith B. *Merrill's atlas of radiographic positioning and procedures*, ed 14, St. Louis, 2019, Elsevier.)



Fig. 15.109 Retrograde urogram—catheter in right ureter, left catheter withdrawn.

Orthopedic Procedures

Orthopedic procedures performed in surgery are intended to reestablish the length, shape, and alignment of fractured bones and joints or to restore function and range of motion of joints affected by trauma or disease. Radiography is required for many orthopedic surgical procedures to provide guidance to the surgeon while reducing fractures, inserting various orthopedic devices, or inserting stabilizing rods in long bones.

Technologists have an important role and responsibility during these procedures. They operate the technology that provides the surgeon with “eyes” or vision during the procedure. C-arm or mobile radiographic units are used extensively during most orthopedic procedures.

ORTHOPEDIC SURGICAL TERMINOLOGY AND CONCEPTS

The following terms, procedures, and concepts are common in orthopedic surgery. Knowledge of these terms is essential because they will be used frequently to describe various orthopedic surgical procedures.

Closed Reduction

Fracture fragments are realigned by manipulation and are immobilized by a cast or splint. A closed reduction is a nonsurgical procedure. The fracture site of the patient is not cut open during the procedure; however, small pins are sometimes placed through the skin of the patient into the proper location and are left in place or later removed.

Open Reduction

For severe fractures with significant displacement or fragmentation, a surgical procedure is required. The fracture site is exposed, and a variety of screws, plates, or rods are inserted as needed to maintain alignment of the bony fragments until new bone growth can take place. This surgical procedure is called an **ORIF**. Images obtained from a C-arm unit or from a mobile radiographic unit are frequently used to guide the orthopedic surgeon during these surgical procedures.

Internal Fixation

During open reduction of fractures, a variety of compression plates, screws, pins, intramedullary rods, nails, or wires are applied to reduce or realign the fracture (Fig. 15.110). Based on the age and condition of the patient, the type of procedure performed, and the extent of the fracture, these devices are left in place and the skin is closed around the devices. In some minor surgeries, these fixation devices may be removed later.

External Fixation

The use of an external fracture-stabilizing device permits bone healing without the immediate requirement for internal fixation. External fixators can also be used in conjunction with internal fixation procedures. Indications for external fixation include severe open fractures, comminuted closed fractures, arthrodesis, infected joints, and major alignment and length deficits. As with internal fixators, several varieties of external fixators are available to aid the surgeon. The Ilizarov device (Fig. 15.111) is a prime example of an external fixator used to correct a length deficit. Through a process of tension stress and distraction, bone length can increase over time through new bone formation. A second external fixation device used for alignment stabilization of the pelvis is shown in Fig. 15.112.

Intramedullary Fixation

Intramedullary rods and nails are inserted within the shaft of long bones to stabilize fractures (Fig. 15.113). This technique is popular for reducing shaft fractures of the humerus, tibia, and femur. In

some cases, the intramedullary rods are a better option than the use of compression plates and screws for reducing midshaft fractures. Intramedullary fixation devices minimize the amount of tissue exposed during surgery, decrease surgical and healing time, and reduce opportunities for postsurgical infection.



Fig. 15.110 Internal fixation devices.

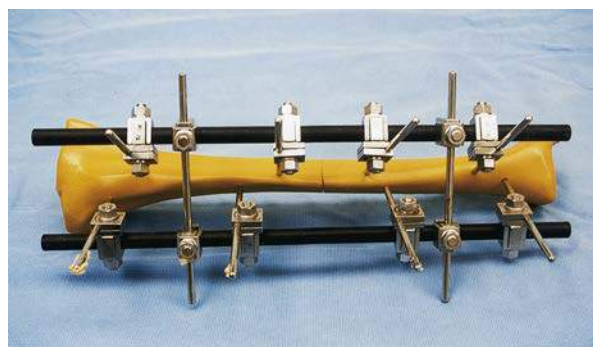


Fig. 15.111 Ilizarov tibial external fixator.

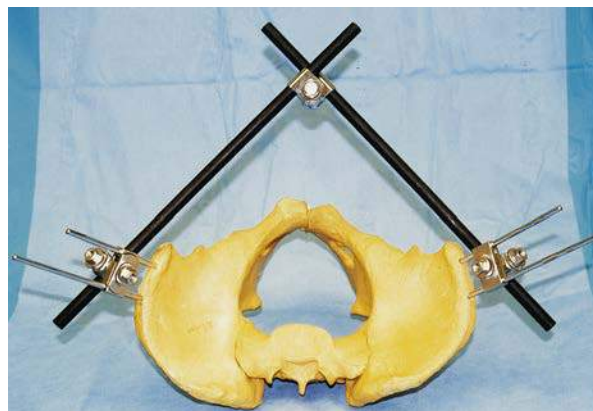


Fig. 15.112 Pelvic external fixator.



Fig. 15.113 Examples of intramedullary fixators—intramedullary rods and nails.

HIP FRACTURES (PINNING, OPEN REDUCTION WITH INTERNAL FIXATION)

Proximal femur (hip) fractures are classified according to their anatomic location. Common hip fractures include **femoral neck** fractures, **intertrochanteric** fractures, and **subtrochanteric** fractures. Each of these fractures can be subclassified (Fig. 15.114). These fractures require an ORIF procedure. Sometimes these ORIF procedures are distinguished by the types of internal fixators used. For a hip pinning, the procedure generally involves the use of long screws for nondisplaced femoral neck fractures (Fig. 15.115).

In a hip ORIF, a longer fixator is placed on the lateral side of the fractured hip and secured by inserting screws through the fixator into the neck and head of the femur, followed by smaller screws below the trochanters that traverse the shaft of the femur. The goals of these surgeries are to reduce and stabilize the fracture and restore use of the lower limb with minimal loss of blood. Internal fixation devices for hip fractures require the use of cannulated screws, compression screws, lag screw and plate combinations, and pins. During the operative procedure, the patient is placed on a special fracture (orthopedic) table that permits traction of the involved limb and fluoroscopy during the procedure (Fig. 15.116).

Based on the type of fracture and fixation device to be used, the fracture first is reduced through traction and manipulation. An incision then is made at the level of the greater trochanter, and the guide pins are inserted through the fracture, thus stabilizing it. For femoral neck fractures, once the guide pins are aligned, large cannulated screws or some other type of internal fixation pin-type device is inserted through the fracture (see Fig. 15.115A to F).

Fluoroscopy is used throughout the procedure to verify the position and location of guide pins and the internal fixator. In some cases, the physician may order a postoperative image of the hip and prosthetic device to verify final alignment of the fracture.

Imaging Equipment Used and Setup

With the fracture table in place, an isolation drape or shower curtain is erected that essentially divides the room into sterile and nonsterile sections and allows easy access and movement of the C-arm outside the sterile field (Fig. 15.117). The C-arm must be free to move easily from a PA to a horizontal beam lateral position. Although the C-arm is located outside the sterile field, a nonsterile bag should cover the x-ray tube (remember that the x-ray tube is located beneath the patient) to prevent blood and povidone-iodine (Betadine) from leaking onto it. The C-arm monitors must be set to allow easy viewing by the surgeon. This generally requires placing the monitors to one side of the surgeon, just behind the shower curtain, where the images can be viewed by the surgeon and technologist without contamination of the sterile area.

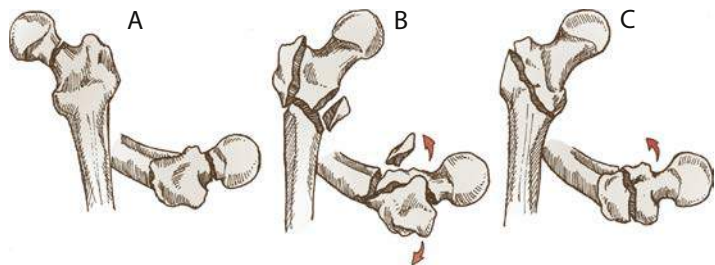


Fig. 15.114 Proximal femur fractures. A, Femoral neck. B, Comminuted subtrochanteric. C, Intertrochanteric. (From Rothrock JC. *Alexander's care of the patient in surgery*, ed 16, St. Louis, 2019, Elsevier.)

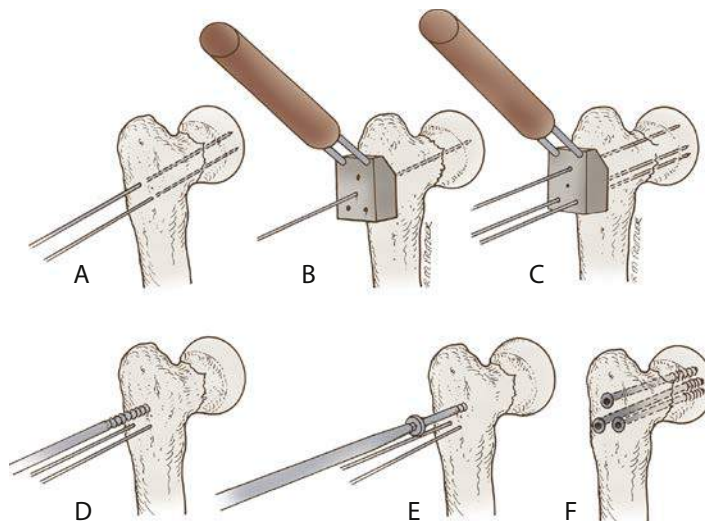


Fig. 15.115 Internal fixation of fractured hip with cannulated screws inserted over guidewires—cannulated screw fixation for nondisplaced femoral neck fractures. (From Rothrock JC. *Alexander's care of the patient in surgery*, ed 16, St. Louis, 2019, Elsevier.)

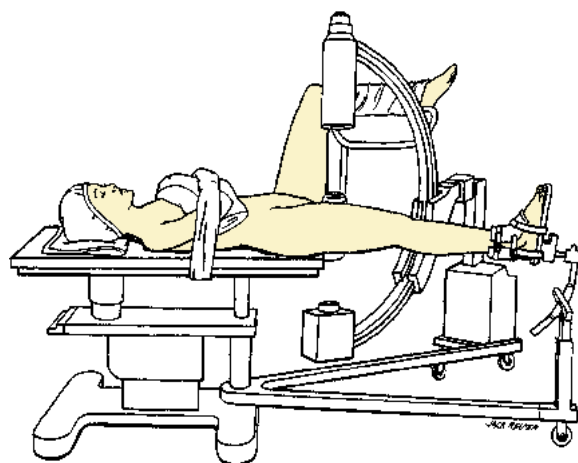


Fig. 15.116 Fracture orthopedic table with C-arm in position. (Adapted from Rothrock JC. *Alexander's care of the patient in surgery*, ed 16, St. Louis, 2019, Elsevier.)



Fig. 15.117 C-arm in PA projection position with intensifier above and tube below, and shower curtain in position separating sterile from nonsterile areas.

Lateral Hip C-Arm Projection

The surgeon will ask the technologist to move the C-arm from its PA position into a lateral hip C-arm projection. This can be accomplished in two ways. One is by sliding the C-arm underneath the affected leg until the x-ray tube is located superiorly and the image intensifier inferiorly (Fig. 15.118). This is not the most desirable manner of placing the C-arm because it increases radiation exposure to the head and neck of the surgeon; however, it may be requested because it is the easiest and quickest way to obtain a lateral hip projection during the surgical case. All images in this projection should have short exposure times, according to the ALARA principle.

The recommended alignment of the C-arm during a lateral projection hip pinning is to place the x-ray tube inferiorly and the image intensifier superiorly and exteriorly above the hip (Fig. 15.119). This alignment will produce the clearest image of the hip while reducing exposure to the head and neck of the surgeon and surgical personnel (see p. 604 in this draft for radiation exposure patterns for the C-arm).

Summary of Procedure

1. The technologist changes into surgical attire and ensures that the mobile unit is functional and clean.
2. After the patient has been sedated and placed on the fracture table, the fracture is reduced and the lower limb is placed in traction to maintain proper alignment of the fracture.
3. Fluoroscopy is used to verify alignment in both PA and lateral perspectives.
4. An incision is made just below the greater trochanter.
5. Guide pins are inserted through the fracture site. The location and position of guide pins are verified with fluoroscopy as needed by the surgeon.
6. A bone reamer is used to provide a channel for a screw or other internal fixator device.
7. A cannulated, lag, or compression screw assembly is inserted over the guide pins. The position of the screws is verified with fluoroscopy in the PA and lateral perspectives.
8. Guide pins are removed and traction is released.
9. The surgical wound is closed.
10. If postoperative radiographs are requested, include the entire orthopedic prosthesis for all projections (Figs. 15.120 and 15.121). This may require an adaptation of positioning principles to ensure that the entire orthopedic prosthesis is included on a single image.

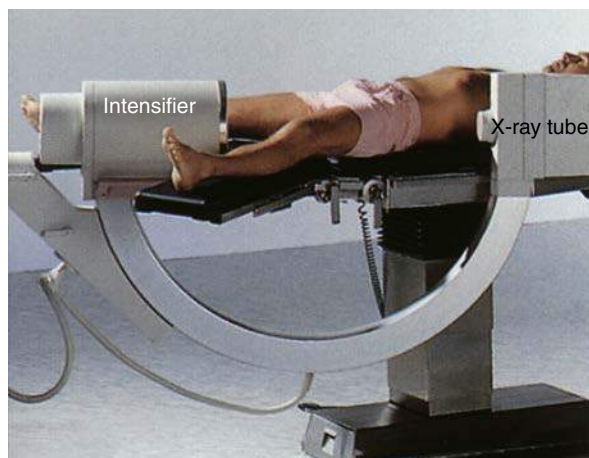


Fig. 15.118 Alternative C-arm alignment if requested by surgeon; not recommended because of increased radiation exposure pattern at tube end. (Courtesy Philips Healthcare, Andover, MA.)



Fig. 15.119 Recommended C-arm alignment for lateral right hip imaging.

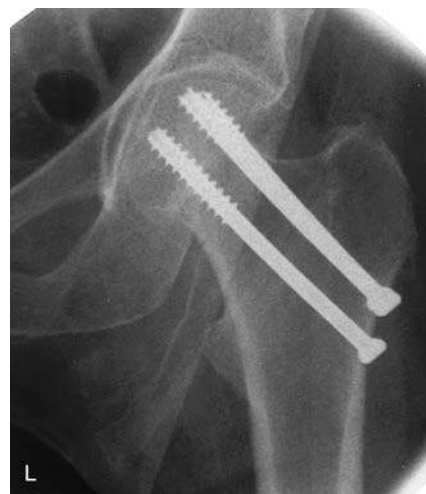


Fig. 15.120 Postoperative radiographic images of hip pinning showing internal fixation devices in place—AP projection. (From Frank E, Long B, Smith B. *Merrill's atlas of radiographic positioning and procedures*, ed 12, St. Louis, 2012, Mosby.)



Fig. 15.121 Postoperative radiographic images of hip pinning showing internal fixation devices in place—lateral projection. (From Frank E, Long B, Smith B. *Merrill's atlas of radiographic positioning and procedures*, ed 12, St. Louis, 2012, Mosby.)

Spinal Procedures

LAMINECTOMY

Laminectomy is a surgical procedure performed to alleviate pain caused by neural impingement. This surgery is designed to remove a small portion of the bone or herniated disk material impinging on the nerve root. The surgery is intended to give the nerve root more space by removing the source of impingement or irritation. Based on the number of vertebrae on which the surgeon operates during the laminectomy, an additional procedure called a **spinal fusion**, which uses rods, plates, and screws to stabilize the surgically altered vertebrae, may be necessary (Figs. 15.122 and 15.123).

Interbody fusion devices, or cages, are another alternative to traditional spinal fusion or the use of pedicle screws to stabilize the vertebrae. **Interbody fusion cages** are titanium cages filled with bone that are inserted between the vertebral bodies to maintain disk space height and fuse the joint, thereby eliminating abnormal movement (Fig. 15.124).

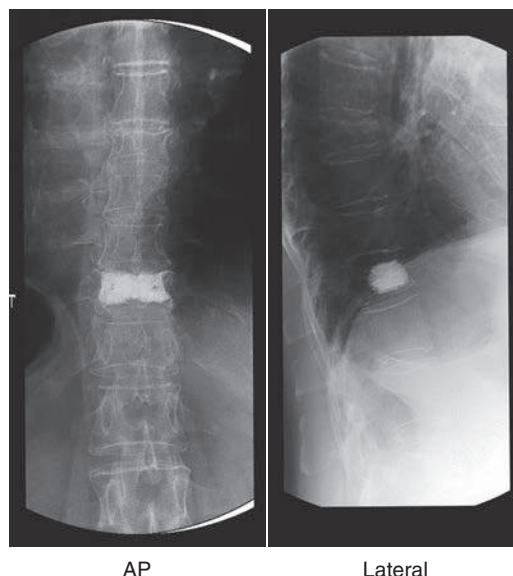
A laminectomy is also effective in decreasing pain and improving function for patients with lumbar spinal stenosis. **Spinal stenosis** is a condition that primarily afflicts older patients; it is caused by degenerative changes that result in enlargement of the facet joints. The enlarged joints then place pressure on the nerves, which may be effectively relieved with a lumbar laminectomy.

A laminectomy can be performed on the cervical or lumbar region. Cervical laminectomy is performed to remove bony obstructions such as bone spurs (osteophytes) and herniated disk materials that cause pain by impinging on the spinal cord or spinal nerves in the cervical region. Lumbar laminectomy is performed for a myriad of reasons, including bony obstructions, stenosis, and spinal cord impingement.

Equipment Used and Setup

The laminectomy may require the use of C-arm or mobile radiographic units. The roles of radiography are to confirm the correct level (or vertebrae) for the laminectomy and provide fluoroscopic guidance if orthopedic plates and/or screws are used during surgery. The C-arm must be free to move easily from an AP or PA to a horizontal beam lateral position (Figs. 15.125 and 15.126). A sterile drape is placed over the image intensifier; a nonsterile bag should cover the x-ray tube to prevent blood and Betadine from leaking onto it. When the x-ray tube is moved to the horizontal

beam lateral position, it may be covered by a sterile drape because this generally places the x-ray tube near the sterile field. C-arm monitors must be set to allow easy viewing by the surgeon.



AP

Lateral

Fig. 15.124 Interbody fusion device—AP and lateral projections.



Fig. 15.125 AP projection (patient prone, tube below) for lumbar laminectomy.



Fig. 15.122 Lumbar laminectomy-fusion—AP projection.



Fig. 15.123 Lumbar laminectomy-fusion—lateral projection.



Fig. 15.126 Lateral projection in position for lumbar laminectomy.

Summary of Procedure

Cervical procedure (anterior approach) The patient is placed in a supine position with the arms drawn down by the sides of the body. The arms may be placed in traction to ensure visibility of the lower cervical vertebrae.

1. The technologist changes into surgical attire and ensures the portable unit or the C-arm is functional and clean.
2. A needle is placed at the level of the laminectomy. The correct vertebral level is verified by C-arm fluoroscopy in both AP and lateral perspectives. A mobile radiographic unit may also be used to confirm correct needle placement by obtaining a horizontal beam lateral projection of the cervical spine. Visualization of the entire spine is necessary for the surgeon to count the number of vertebrae correctly and determine the correct level (vertebral) at which the procedure should be performed.
3. The C-arm must be parallel to the vertebrae to avoid distortion of visible structures.
4. Cervical plates and screws may be used during the procedure to stabilize the vertebrae. C-arm fluoroscopy may be used as needed to guide placement of the orthopedic appliances.

Lumbar procedure (posterior approach) The patient is prone with a bolster under the abdomen to flex the spine. The arms usually are above the head on arm boards.

1. The technologist changes into surgical attire and ensures that the portable unit or C-arm is functional and clean.

2. A needle is placed at the level of the laminectomy. The correct vertebral level is verified by C-arm fluoroscopy in both AP and lateral perspectives. A mobile radiographic unit may also be used to confirm correct needle placement by obtaining a horizontal beam lateral projection of the lumbar spine. Visualization of the entire spine is necessary for the surgeon to correctly count the number of vertebrae and determine the correct level (vertebral) at which the procedure should be performed.
3. The C-arm must be parallel to the vertebrae to avoid distortion of visible structures.
4. Pedicle screws, interbody fusion cages, rods, and other appliances may be used during the procedure. C-arm fluoroscopy may be used as needed to assist with the placement of orthopedic appliances.

Anatomy demonstrated The spine in the PA-AP and lateral perspectives at the desired level within the entire vertebral (cervical or lumbar) column is seen. The total vertebrae, including the spinous processes, must be demonstrated.

Thoracic Procedures

PACEMAKER INSERTION

More than 500,000 Americans have an implantable permanent **pacemaker** device. A pacemaker implantation is performed under local or general anesthesia in a hospital by a surgeon assisted by a cardiologist. An insulated wire called a *lead* is inserted into an incision below the clavicle and is guided through a large vein into the chambers of the heart. These electrodes stimulate the heart muscle, causing it to beat at a predetermined rate. This process is referred to as *pacing* of the heart.

Electrodes often are inserted through a vein in the arm or chest and are advanced into the right ventricle under fluoroscopic guidance. The pulse generator or battery that provides the electrical sensation to control the heartbeat may be external (temporary) or may be inserted into the superficial tissues of the thoracic wall.

Equipment Used and Setup

A pacemaker or other line insertion procedure can be performed in the surgical suite, outpatient surgery suite, or radiology department. The use of fluoroscopy is essential during the insertion of electrodes into the right ventricle of the heart.

C-arm mobile fluoroscopy is used in the surgical suite (Fig. 15.127). The C-arm must be free to move easily from a PA to a horizontal beam lateral position. A sterile drape is placed over the image intensifier. C-arm monitors must be set to allow easy viewing by the cardiologist or surgeon.

Summary of Procedure—Transvenous Approach

1. The technologist changes into surgical attire and ensures that the C-arm is functional and clean.
2. Venotomy is performed (vein is accessed).
3. Under fluoroscopic guidance, the electrode is advanced through the vein into the right atrium, through the tricuspid valve, and into the right **ventricle**. The tip of the electrode is advanced until it reaches the right ventricular apex.
4. A pulse generator is inserted into the chest wall.

Once the procedure is complete, the patient's vital signs are monitored.

A postoperative AP or PA chest x-ray is taken to ensure that the pacemaker and leads are properly positioned. In this example, the patient must be positioned as upright as possible to confirm proper positioning of the lead wires (Figs. 15.128 and 15.129).



Fig. 15.128 PA chest with pacemaker in place. (From Jaroszewski DE et. al. Nontraditional surgical approaches for implantation of pacemaker and cardioverter defibrillator systems in patients with limited venous access. *The Annals of Thoracic Surgery* 88(1):112–116.)



Fig. 15.129 Lateral chest with pacemaker in place. (From Jaroszewski DE et. al. Nontraditional surgical approaches for implantation of pacemaker and cardioverter defibrillator systems in patients with limited venous access. *The Annals of Thoracic Surgery* 88(1):112–116.)



Fig. 15.127 Radiography in the surgical suite.

Glossary of Surgical Abbreviations, Terminology, and Procedures

ACL Anterior cruciate ligament

Arthrodesis (*ar-thro-de'-sis*) Stiffening of a joint by operative means

Arthropathy (*ar'-throp-a-the*) Any disease that affects a joint

Arthroplasty (*ar'-thro-plas-te*) Creation of an artificial joint to correct ankylosis

Asepsis (*a-sep'-sis*) A state of sterility; condition in which living pathogens are absent

Cancellous screw Orthopedic screw designed to enter and fix porous and spongy bone

Cannulated screw Large screw used for internal fixation of non-displaced fractures of proximal femur

Cardiac pacemaker An artificial regulator for cardiac rate and rhythm

Cerclage wire Orthopedic wire that tightens around fracture site to reduce shortening of limb

Cesium (*se'-ze-um*) **implants** The use of radioactive cesium in the treatment of certain malignancies, including prostate cancer

Cholecystectomy (*ko'-le-sis-tek'-to-me*) Surgical removal of gallbladder

Closed reduction Procedure in which bone fragments are reduced manually without surgical intervention

Cortical screw Narrow orthopedic screw designed to enter and fix cortical bone

CR Closed reduction (cast or traction)

Cystoscope (*sis'-to-skop*) Lighted tubular endoscope used for examination of the urinary bladder

DHS Dynamic hip screw

Dynamic compression plate Screw and plate combination used to apply forces through the fracture site; used commonly for long bone shaft fractures in which stress may be great

ESWL (extracorporeal shock wave lithotripsy) Electrohydraulic shock waves used to break apart calcifications in the urinary system

EX-FIX External fixation

Fracture (orthopedic) table A special OR table used for hip pinning and other orthopedic procedures to provide traction to the involved limb and allow fluoroscopy to be performed during the procedure

Hip pinning Surgical procedure designed to reduce proximal femoral fractures through the use of various internal fixation devices

HTO High tibial osteotomy

Ilizarov technique Procedure in which a special external fixator is used to lengthen long bones as treatment for severe fracture or congenital deformity

IM nail Intramedullary nail

Interbody bone fusion device Titanium or other alloy cage filled with bone and inserted between the vertebral bodies to maintain disk space height and permit fusion of the intervertebral joint

Intramedullary rod A flexible or rigid device placed within the medullary cavity to reduce a fracture or stabilize a diseased long bone

Kirschner (*kirsh'-ner*) **wire (K-wire)** Unthreaded (smooth) or threaded metallic wire used to reduce fractures of the wrist (carpals) and individual bones of the hands and feet; also may be used for skeletal traction

Laminectomy (*lam'-i-nek'-to-me*) A surgical procedure performed to alleviate pain caused by neural impingement by removing an aspect of the lamina in the vertebral arch

Laminotomy (*lam'-i-not'-o-me*) Surgical opening into one or more laminae of the vertebral arch

Laparoscopic (*lap'-ah-ro-skop'-ik*) **cholecystectomy** Use of a special endoscopic device to visualize and assist with surgical removal of the gallbladder

Lithotripsy (*lith'-o-trip-se*) Crushing of calcification in the renal pelvis, ureter, or urinary bladder by mechanical force or sound waves

Microdiskectomy (*mi'-kro-dis-kek'-to-me*) Microsurgical procedure performed on the spine to remove bony fragments or disk material that may be causing neural impingement

Neural impingement A condition in which bony changes or a herniated disk produces impingement of the spinal nerves that pass through the vertebral arch of the vertebra

Open reduction of fracture fragments through surgical intervention

Operative (immediate) cholangiography Radiographic procedure performed during surgery to visualize and locate undetected stones or obstructions within the biliary ducts

ORIF Open reduction with internal fixation

PCL Posterior cruciate ligament

Prosthesis (*pros-the'-sis*) Fabricated (artificial) substitute for a diseased or missing anatomic part

Reduce To align two bone fragments in the correct position as treatment for a fracture, as applied in orthopedic medicine

Retrograde urography A nonfunctional examination of the urinary system during which contrast medium is introduced directly retrograde (backward, against the flow) into the pelvicalyceal system via catheterization by a urologist during a minor surgical procedure

Semitubular plate Flexible and thin orthopedic plate used to fix and connect fractures

Shower curtain An isolation drape that separates the sterile field from the nonsterile environment; often used to permit the use of C-arm fluoroscopy during a hip pinning procedure

Spinal fusion Surgical fusion of one vertebra to another, which stabilizes them following laminectomy or as treatment for a degenerative condition or fracture

Spinal stenosis Condition caused by degenerative changes that result in enlargement of the facet joints, which often leads to impingement of the spinal nerves that pass by them

Strike-through Soaking of moisture through a sterile or nonsterile drape, cover, or protective barrier, permitting bacteria to reach sterile areas

THR, THA Total hip replacement, total hip appliance

TKR, TKA Total knee replacement, total knee appliance

Total joint arthroplasty The use of artificial joint implants to restore motion and function of a joint—for example, total hip replacement is common orthopedic procedure performed on patients with degenerative joint disease (e.g., avascular necrosis [AVN] of proximal femur)

Traction The process of putting a limb, bone, or group of muscles under tension with the use of weights and pulleys to align or immobilize the part

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